

Maintenance Support Performance Analysis (MSPA)

23-25 July 2024

Zenith Hotel, Kuantan

Trainers:

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Learning Objectives

At the end of the training, participants should:

1. Understand the factors affecting maintenance process in Plant environment and the relationship between maintenance support and RAM
2. Understand the component/elements that drive the maintenance activities and task execution for both planned and unplanned work
3. Understand key performance indicators and resource optimization through lean concepts and tools
4. Understand the material supply chain, inventory management and economic order quantity and lead time
5. Be able to perform MSPA process
6. Be able to recommend improvement for MSP gaps closure

Training Agenda



Start Time	End Time	Day 1	Day 2	Day 3
9:00 am	10:30 am	Introduction Pre-Test Overview of Maintenance and Maintenance Support	Statistical Data Analysis for Maintenance	Lean maintenance
10:30 am	10:45 am	Tea Break		
10:45 am	12:30 pm	Overview of Maintenance and Maintenance Support (ct'd)	Statistical Data Analysis for Maintenance	MSPA with Simulation
12:30 pm	2:00 pm	Lunch/Prayer		
2:00 pm	3:30 pm	RAM And Maintenance support, Data Collection in Maintenance	Materials and Spare Part Management in Maintenance	Maintenance Planning and Scheduling
3:30 pm	3:45 pm	Tea Break		
3:45 pm	5:00 pm	Data Collection in Maintenance (ct'd)	Lean maintenance	Conclusion and Post Test

Ice-breaking

- Name
- Job function / Position / Area
- Any experience in maintenance and reliability analysis?
- What do you expect from this course?



Overview of Maintenance and Maintenance Support

What is Maintenance?

“Combination of all technical and management actions intended to retain an item in, or restore it to, a state in which it can perform as required”

ISO 14224

“Combination of any actions carried out to retain an item in or restore it to, conditions in which it is able to meet the requirements of the relevant specification and perform its required functions”

IEC 61892-7

Why perform Maintenance?

- Maintenance in the oil and gas industry is a critical activity that ensures the safe and efficient operation of equipment and facilities
- Proper maintenance is vital to prevent accidents, extend equipment life, and minimize operational downtime.



Types of maintenance

- I. Reactive Maintenance
- II. Preventive Maintenance (PM)
- III. Condition-Based Maintenance (CBM)
- IV. Predictive Maintenance (PdM)
- V. Prescriptive Maintenance

Reactive Maintenance



Reactive maintenance is a maintenance strategy where equipment is only repaired **once it has broken down / malfunctioned**, to restore the equipment to normal operating condition

Can be a strategy when the equipment/machine:

- Downtime is not costly (time and money)*
- Easily Fixable*
- Replacement/spare parts are readily available*
- There are redundant equipment*

Pros

- No/lower upfront cost or cost to manage
- Minimal maintenance staff needed

Cons

- Unpredictable and inconsistent downtime
- High costs associated with fixing a failed machine
- Repairs tend to be urgent which may require overtime labor
- Shorter equipment life expectancy
- Higher energy costs

Types of Reactive Maintenance

1. Breakdown Maintenance

Action taken on a piece of equipment that won't start or operate whatsoever. The machinery is completely broken, and may require extensive repairs to run again (or may need to be replaced entirely).

In most cases, breakdown maintenance is an **unplanned event**, and the repairs are urgently requested in order to resume normal business operations

2. Run-to-Failure Maintenance

Deliberately allowing a machine to run until it breaks down. After the failure, reactive maintenance is performed—no prior or preventive maintenance is performed on the machine in advance.

Run-to-fail is always a deliberate decision (i.e. planned), and there is some type of plan in place to repair the item without interrupting production. In addition, the equipment that is allowed to break down does not pose a health or safety risk to operating personnel.

3. Corrective Maintenance

Actions that target and repair a system malfunction so that equipment can be restored to proper working order. One plus of corrective maintenance is that the defect may be caught or noticed before it causes a significant problem, or results in a total equipment breakdown.

Corrective maintenance is that it does not require the repair of a totally-busted asset. Rather, corrective maintenance is performed to correct a defect, or a potential problem area, in equipment that is still running. This gives the business more time to respond to issues, and even mimics preventive maintenance by allowing for the opportunity to correct problems before they escalate.

4. Emergency Maintenance

Emergency maintenance requires a last-minute response to the sudden breakdown of an asset. However, emergency maintenance entails some type of threat to health and safety (e.g. safety locks fail on a machine with hazardous moving parts).

Preventive Maintenance (PM)



A **proactive regular and routine** (time or usage based) maintenance of equipment and assets in order to keep them running and prevent any costly unplanned downtime from unexpected equipment failure.

PM aims to prevent any deterioration in the machine from occurring in the first place. It is easier to plan and work around and it's also generally less expensive to replace a simple component a few times than repairing a damaged machine after failure.

Pros

- Improve reliability
- Increase asset life span
- Reduce unplanned downtime
- Reduce operation errors and HSE risks
- More efficient use of energy and resources

Cons

- Schedule may be too frequent (wasteful)
- Schedule may be too infrequent (misses a potential failure)
- Can't catch uncommon failure modes
- Budget / resources / time constraints

Condition Based Maintenance (CBM)



A strategy that **monitors the real-time condition** of an asset performance to decide what maintenance needs to be performed.

CBM aims to monitor and spot upcoming equipment failure or decreasing performance, so maintenance can be proactively scheduled when needed—and not before. Asset conditions need to trigger maintenance within a long enough time before failure, so work can be finished before the asset fails or performance falls below the optimal level.

Pros

- Increases asset life span
- Real-time (and often remote) indication of the health of an asset
- Maintenance only performed when needed
- Advanced warning provided to allow for convenient scheduling of maintenance

Cons

- Requires monitoring hardware and software to generate warnings which have upfront and recurring costs
- Some software may require the user to be specially trained to operate

CBM (ct'd)

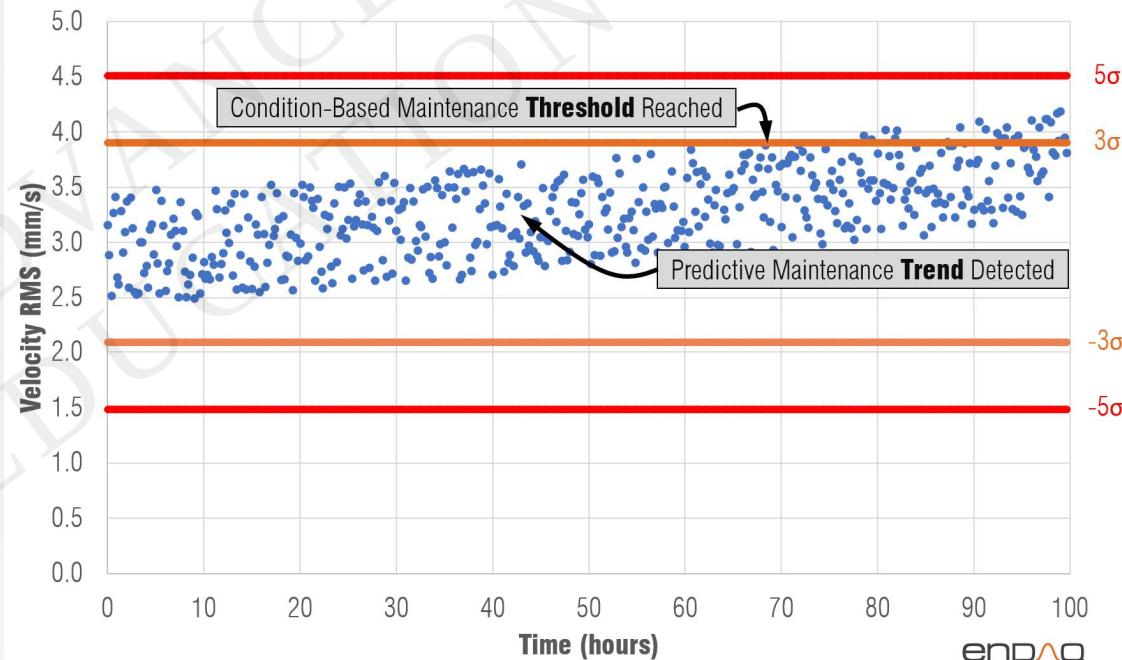
CBM uses sensors mounted to a machine to periodically check the health of that asset. Then it compares the most recently measured value against preset "conditions" to be reached which signify an alarm or warning. Typically this is done by:

1. Establishing a baseline of a known good or healthy machine
2. Calculate a normal distribution of the data taken
3. Set warning levels at ± 3 standard deviations and then alarms at ± 5 standard deviations off the baseline levels

Many condition-based maintenance systems require the user to define warning and alarm thresholds. Some "advanced" ones will automatically do that baseline determination for a period of time and then calculates a simple normal distribution to set warnings & alarms using the standard deviation



Vibration Condition Monitoring Values Over Time



Predictive Maintenance (PdM)



The strategy that uses **data-driven**, proactive maintenance methods that are designed to analyze the condition of equipment and help **predict when machine will fail**, and maintenance should be performed

PdM aims to predict when equipment failure might occur and to prevent its occurrence by performing maintenance

Pros

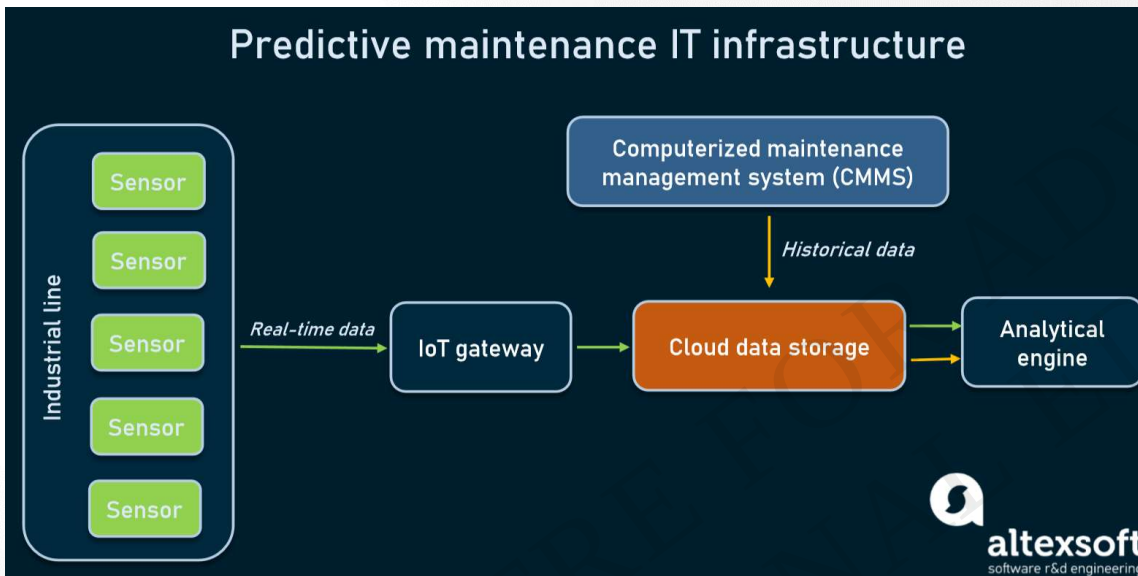
- Increases asset life span
- Real-time indication of the health of an asset
- Maintenance only performed when needed
- Advanced warning provided to allow for convenient scheduling of maintenance

Cons

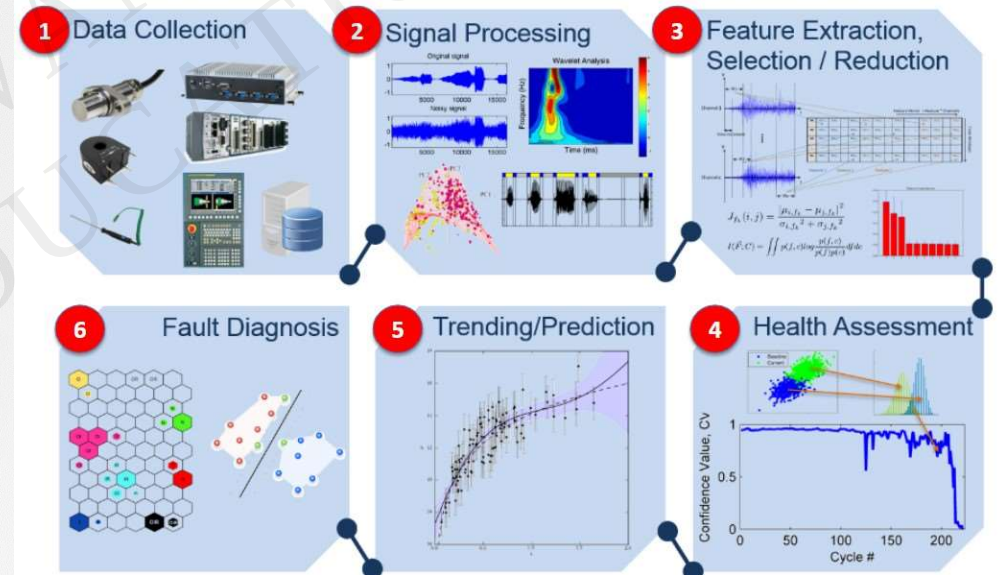
- Requires lots of data collection
- Requires monitoring hardware and software to generate warnings which have upfront and recurring costs
- Some software may require the user to be specially trained to operate

Predictive Maintenance (PdM) ct'd

Predictive maintenance IT infrastructure



Predictive Template Deployment



Prescriptive Maintenance



Prescriptive maintenance uses more data to tell **WHAT SPECIFIC PART will fail** in the machine and how to **fix** the problem

In order to have a prescriptive maintenance program, you need to be able to correlate specific signatures in the raw data and trends to specific root causes.

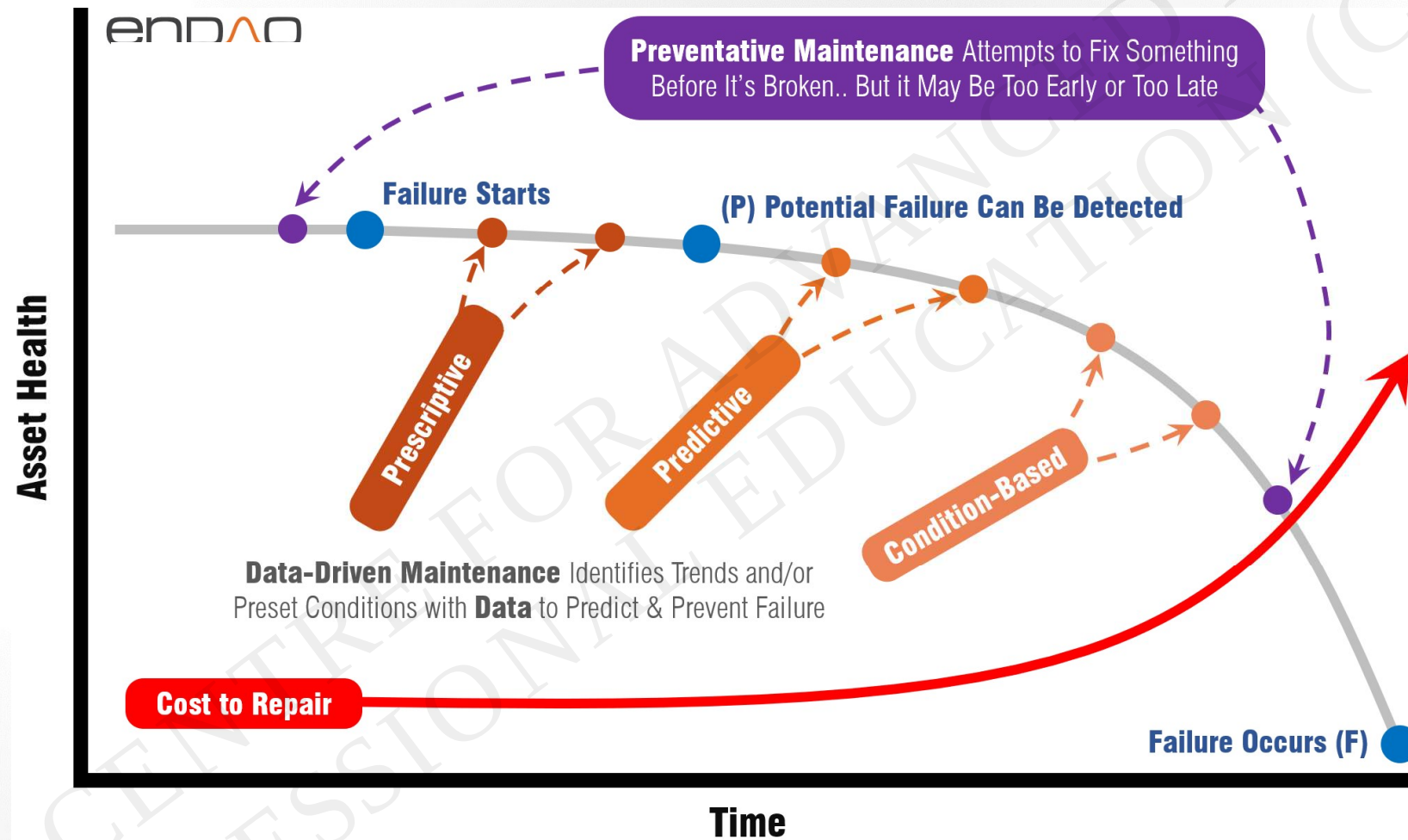
Pros

- Increases asset reliability and life span
- Specific action/maintenance can be performed on specific part/problem, rather than in general

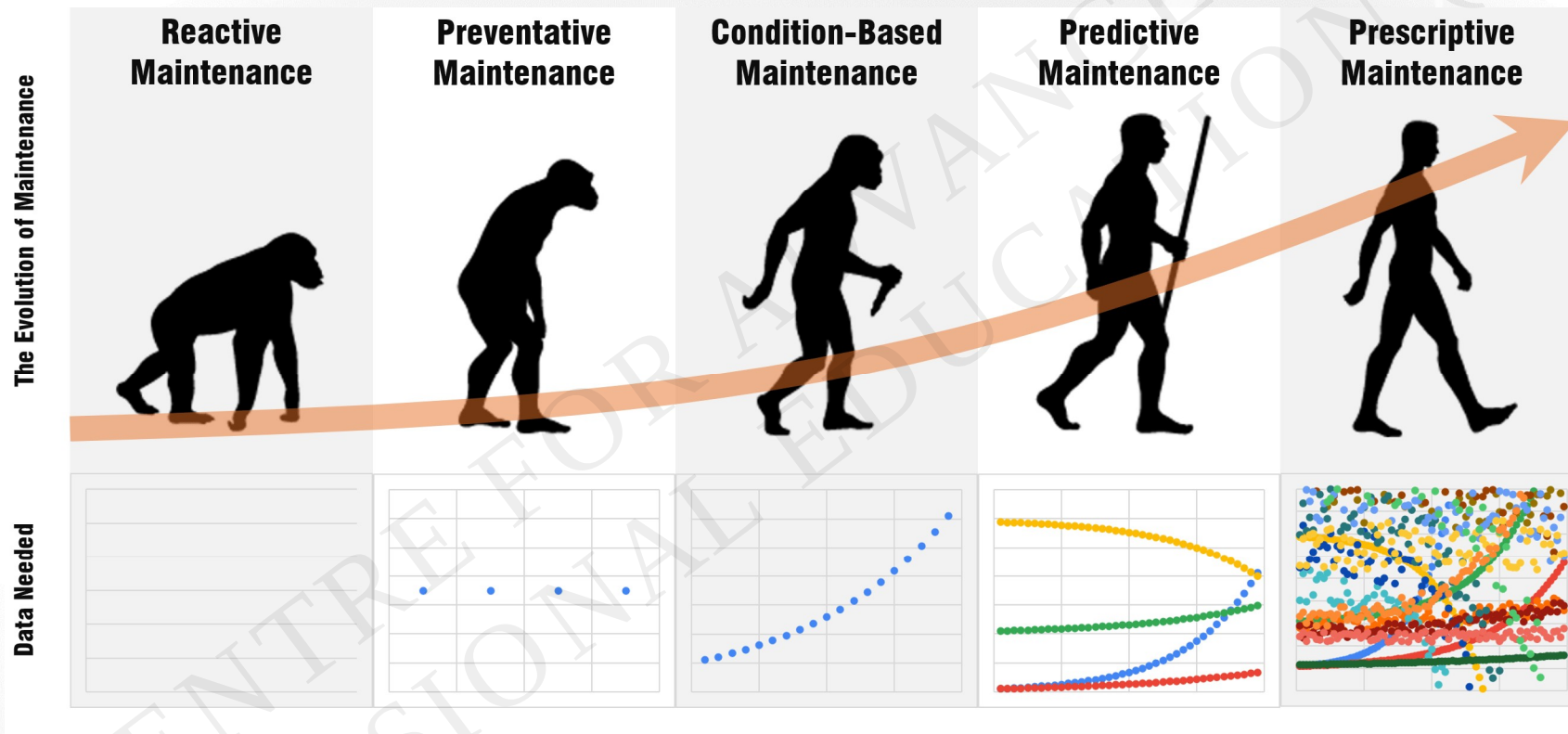
Cons

- Requires monitoring hardware and software which have upfront and recurring costs
- Requires lots of data and analysis

P-F Curve vs. Different Types of Maintenance



Data driven Maintenance - Evolution



Maintenance Performance Evaluation

- Assessment of maintenance performance is crucial for any organization to identify areas of improvement and make informed decisions to optimize their maintenance processes.
- With proper assessment organizations can understand the impact of maintenance on overall operations, equipment reliability and cost management
- Provide insights into the effectiveness of maintenance strategies, allowing organizations to make data-driven decisions for continuous improvement
- Assist on prioritizing maintenance activities based on their significance and criticality, allocating resources effectively, reducing downtime, and maximizing equipment availability

Maintenance Performance Key Metrics

- To assess maintenance performance accurately, key metrics / Key performance indicators (KPIs) that align with maintenance objectives and goals need to be identified.
- Common KPIs for assessing maintenance performance:
 - equipment uptime
 - mean time between failures (MTBF)
 - mean time to repair (MTTR)
 - planned maintenance percentage
 - emergency maintenance percentage
 - maintenance backlog

Maintenance Performance Key Metrics

- By tracking these metrics, organizations can
 - ✓ gain insights into the Maintainability, Reliability, Availability, performance of assets.
 - ✓ help in identifying patterns, trends, potential issues that impact maintenance performance.
 - ✓ basis for benchmarking, setting performance targets
 - ✓ enabling organizations to monitor progress and drive continuous improvement.

Example Key Metric : Maintainability (MTTR)

- At the highest level, maintainability can be seen as a product of overall support programme in the system where high maintainability reflects the effectiveness of in the design approach, manpower allocations, training delivery and supply chain management (*Knezevic, 1997*).

Concept of Maintenance and Maintenance Support (IEC Standard 60300-3-14)

❑ Definition: Maintenance

- Combination of **all technical and administrative actions**, including supervision actions, intended to retain an item in, or restore it to a state in which it can perform a required function.

❑ Definition: Maintenance Support

- Resources required to maintain an item under a given maintenance concept and guided by a maintenance policy.

❑ Definition: Maintenance Concept

- Interrelationship between the maintenance echelons, the indenture levels and the levels of maintenance to be applied for the maintenance of an item

❑ Definition: Maintainability (performance)

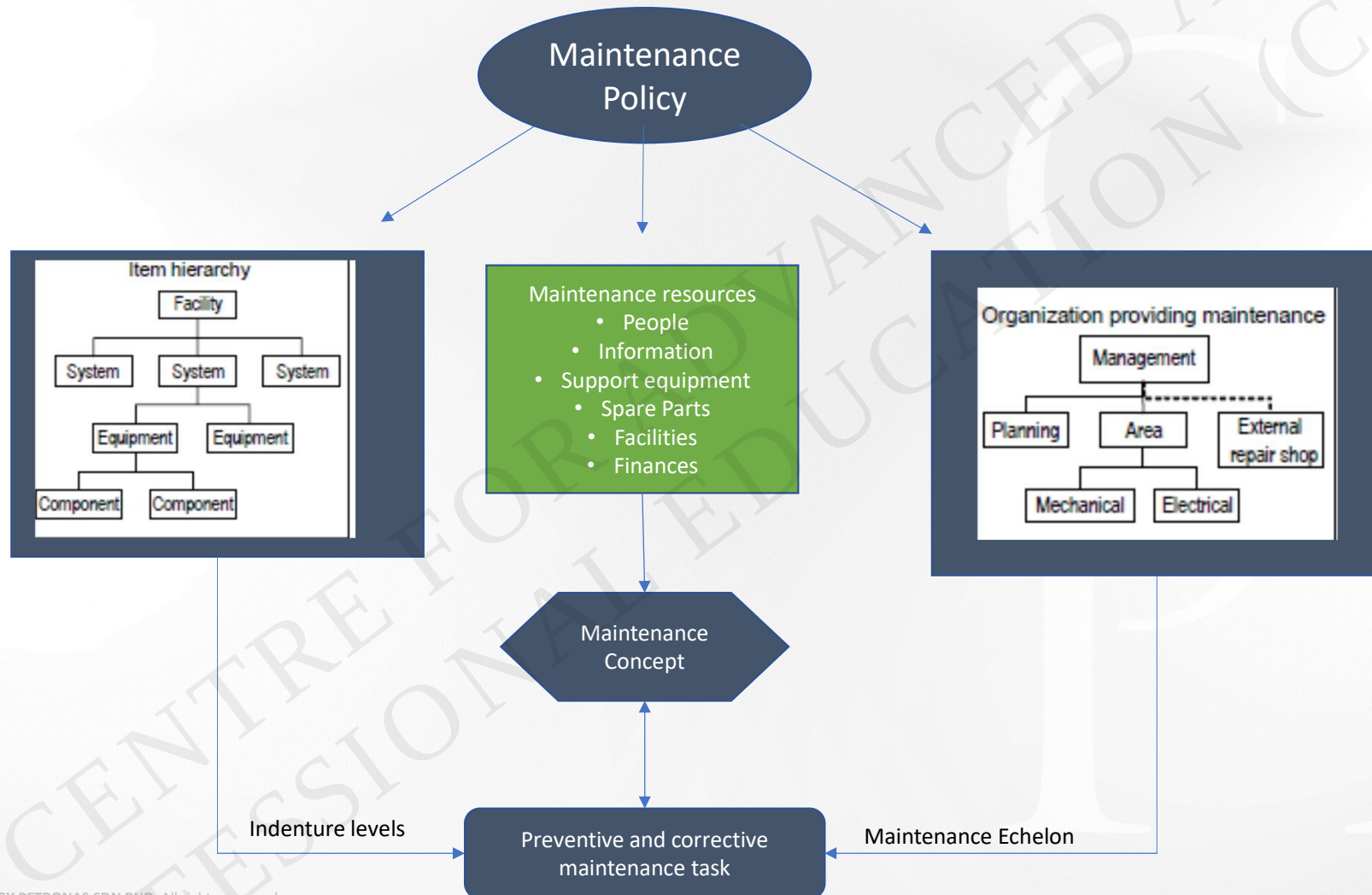
- ability of an item under given conditions of use, to be retained in, or restored to, a state in which it can perform a required function, when maintenance is performed under given conditions and using stated procedures and resources

NOTE: The term maintainability is also used as a measure of maintainability performance

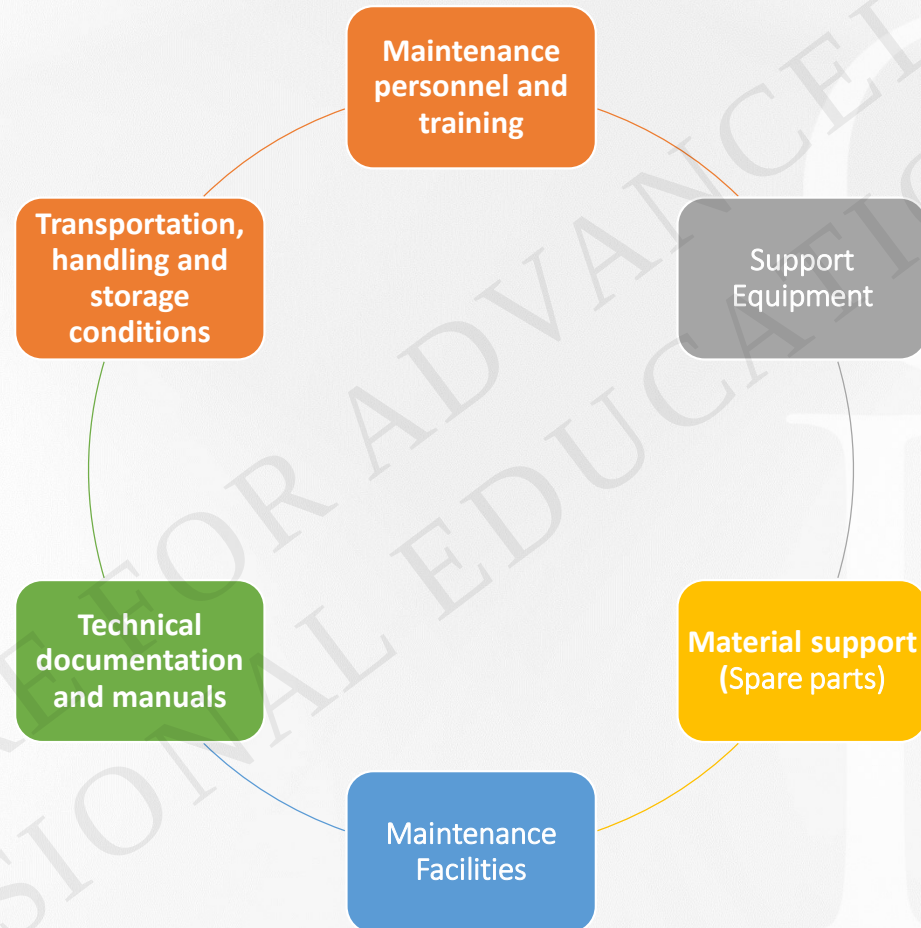
Concept of Maintenance and Maintenance Support

- Definition: Maintenance Policy
 - General approach to the provision of maintenance and maintenance support based on the objectives and policies of owners, users and customers
- **Definition: Maintenance Support Performance**
 - Ability of a maintenance organization, under given conditions, to provide upon demand, resources required to maintain an item, under a given maintenance concept and guided by a maintenance policy.
NOTE: The given conditions are related to the item itself and to the condition under which the item is used and maintained.
- Definition: Maintenance Echelon
 - Position in an organization specified levels of maintenance are to be carried out on an item
NOTE 1 : Examples of maintenance are field, repair shop, manufacturer
NOTE 2 : The maintenance echelon is characterized by the level of skill of the personnel, the location, etc.

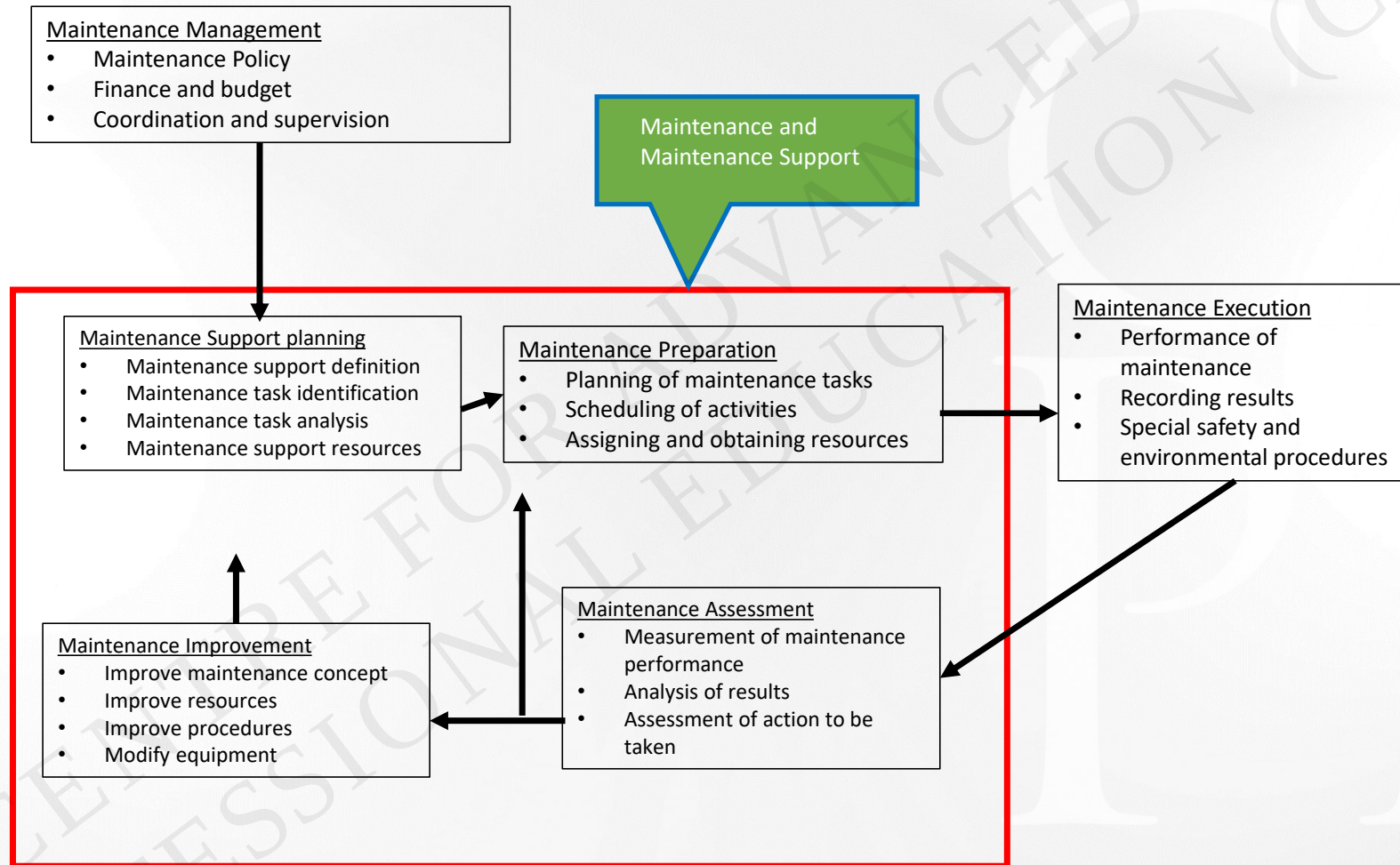
Interrelationship of Maintenance terms



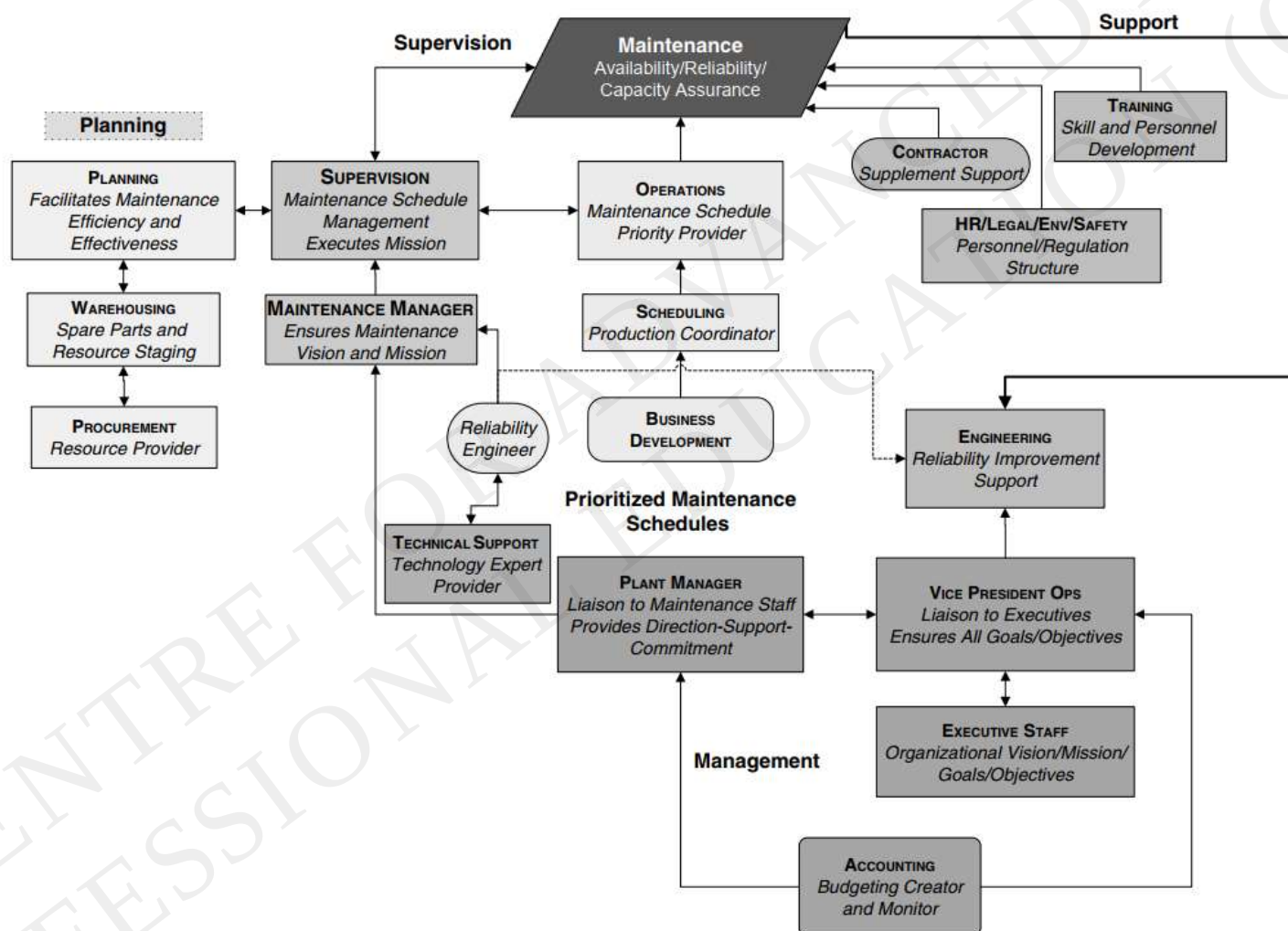
Maintenance Support Elements



Maintenance Management

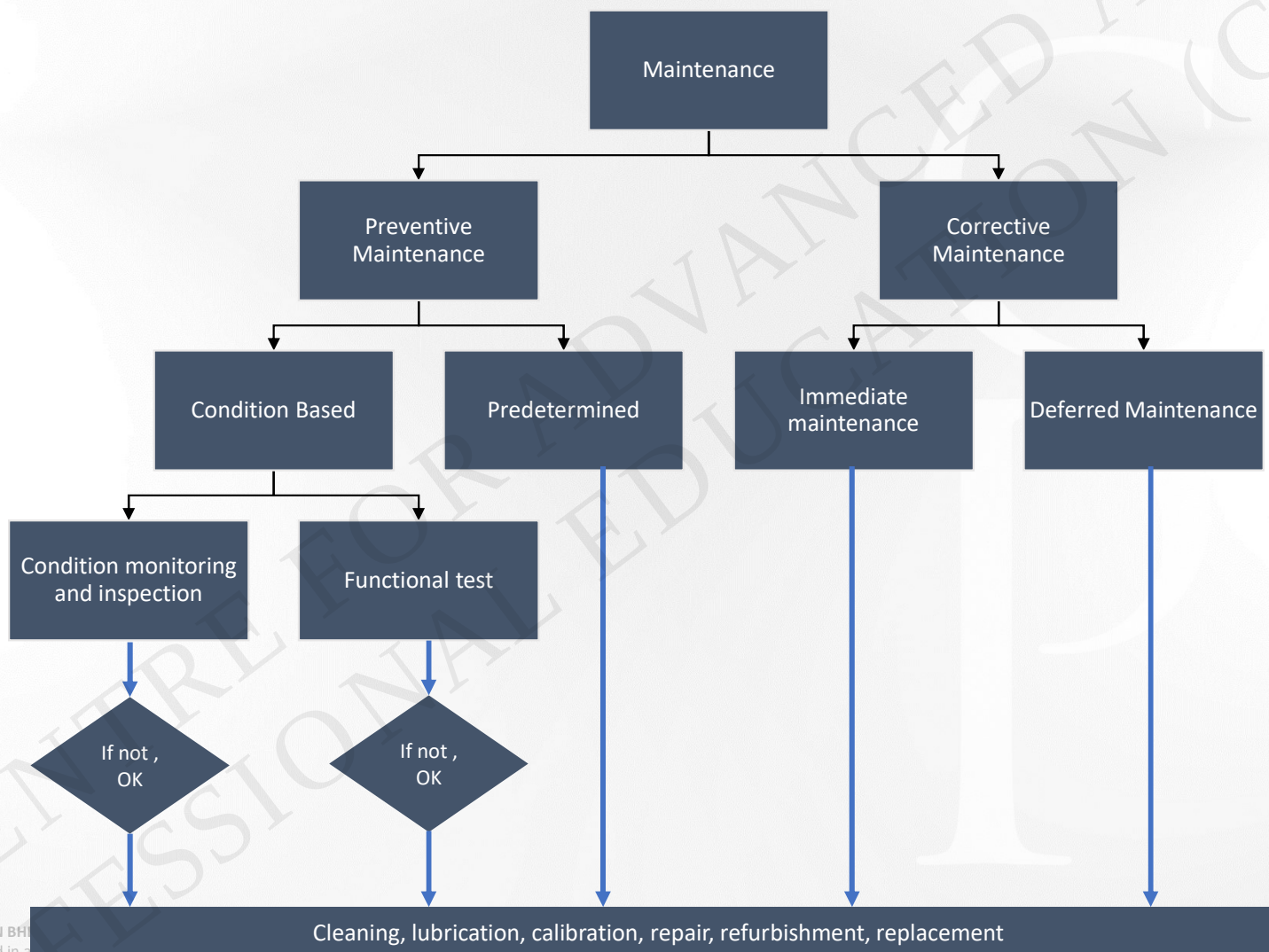


Functional maintenance organizational structure

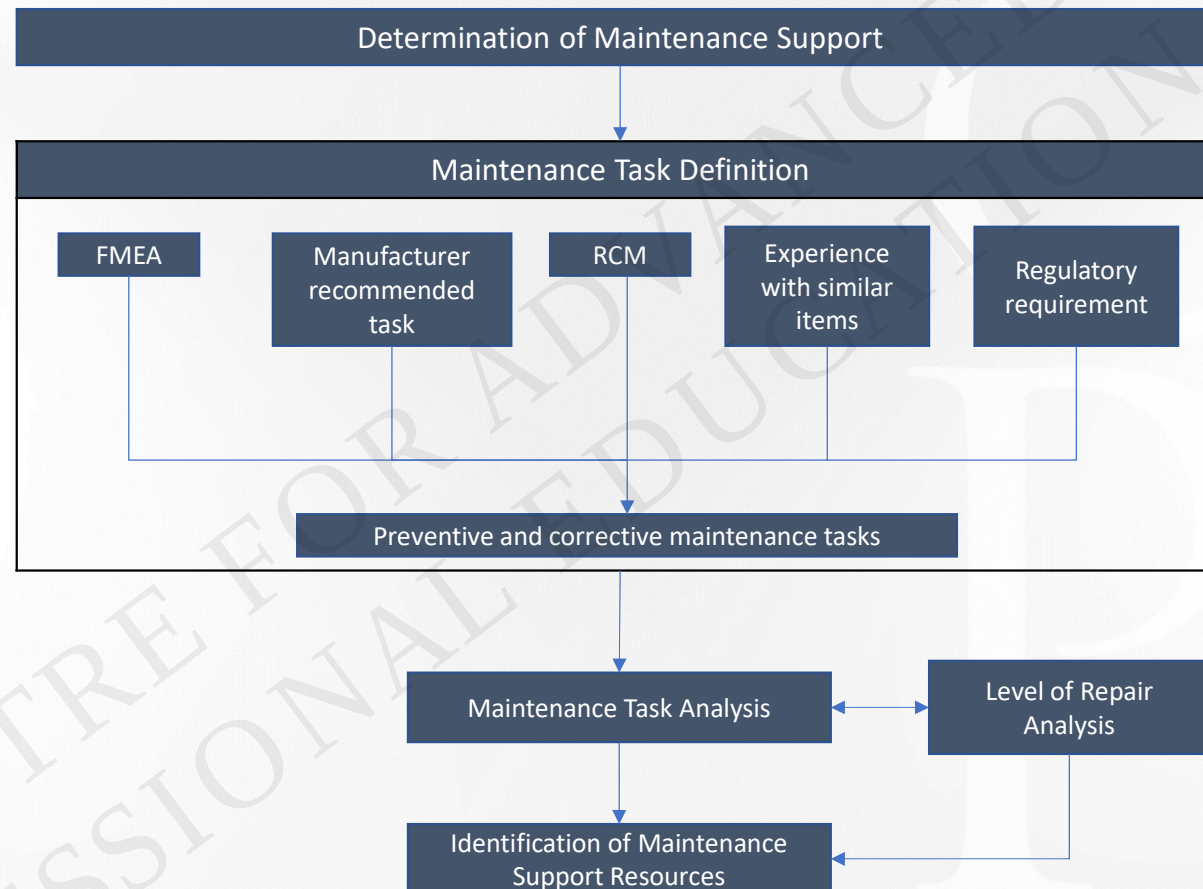


Rules of thumbs for maintenance and reliability engineers; R. Smith, RK Mobley

Types of maintenance tasks – IEC 262/04



Maintenance and maintenance support planning process – IEC 264/04



Maintenance Task Analysis

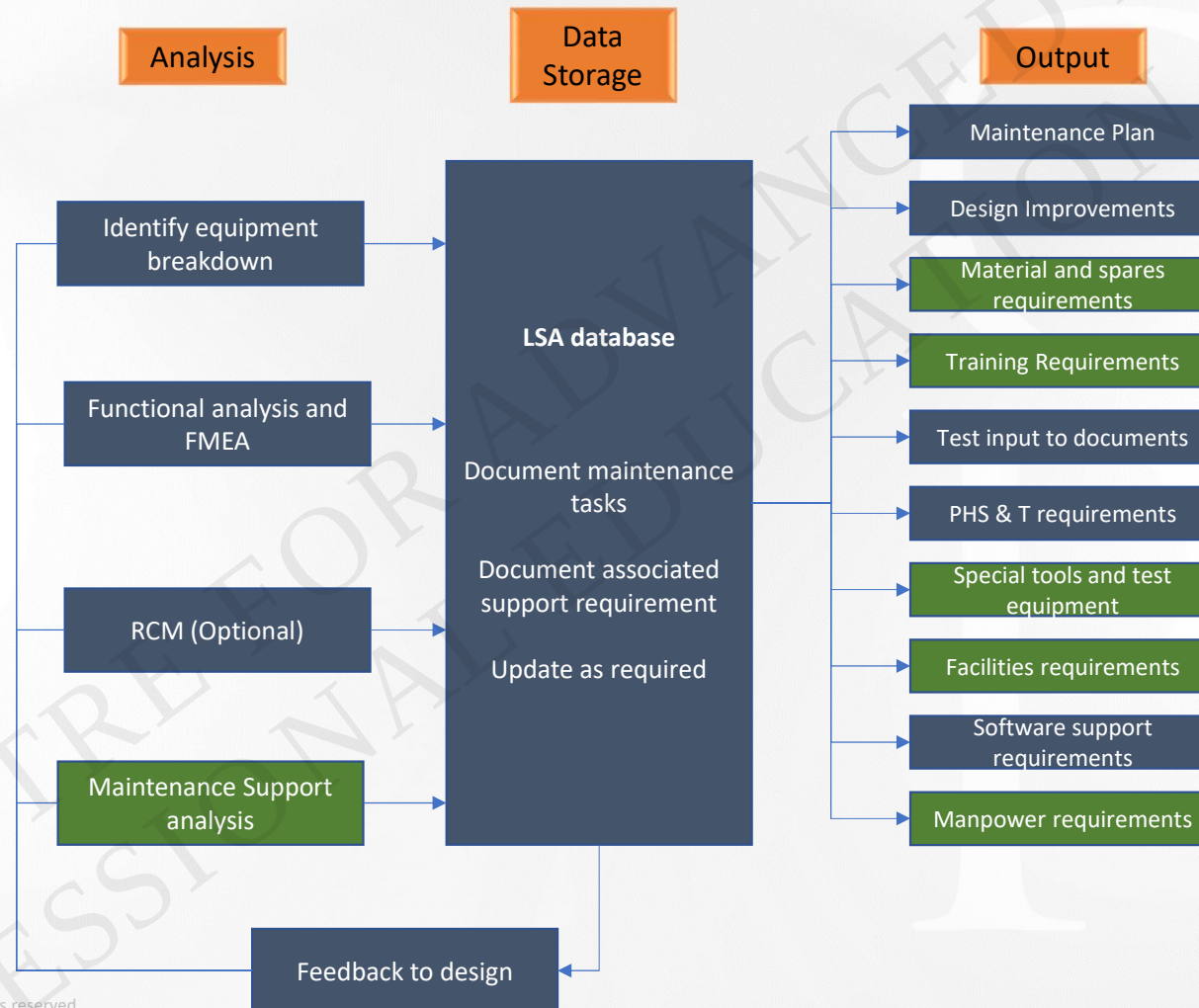
- Maintenance **task analysis is concerned with the resources needed to accomplish maintenance**. Typically, it forms part of the maintenance planning process which preceded operational deployment. Maintenance task analysis is an essential part of the integrated logistic support procedure described in IEC 60300-3-12 and is also discussed in IEC 60300-3-14.
- A review of the maintenance task analysis may be used for qualitative verification.
- Maintenance Support Analysis (MSA) is used to detail the operation and maintenance tasks required for the product and RCM may be used in the analysis. Its purpose using a step-by-step review of each maintenance task, is to identify the **manpower, skills, tools, test equipment, resource or spares** necessary to carry out the tasks.

Example: Nascar Pit Stop

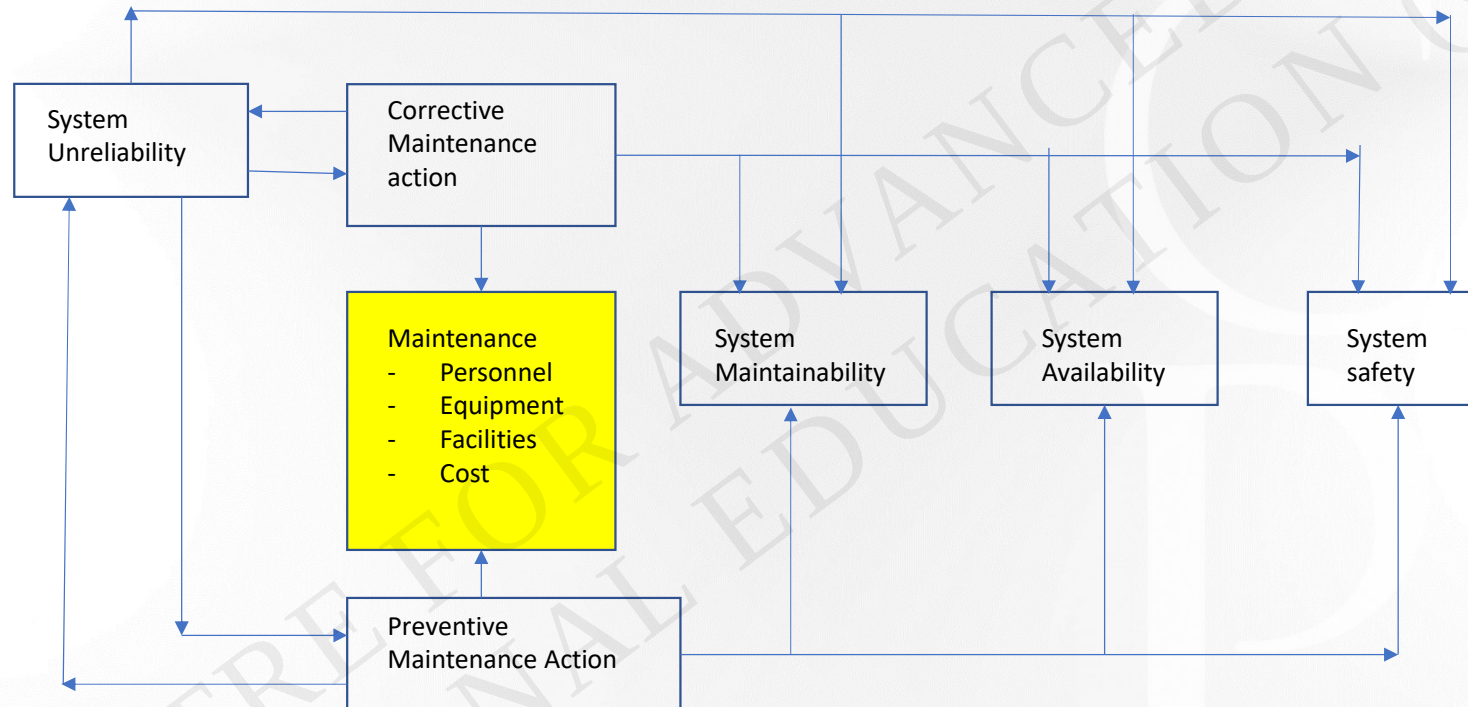


<https://www.youtube.com/watch?v=o7J1ICj1zeE>

Maintenance Support Analysis (IEC 2517/01)

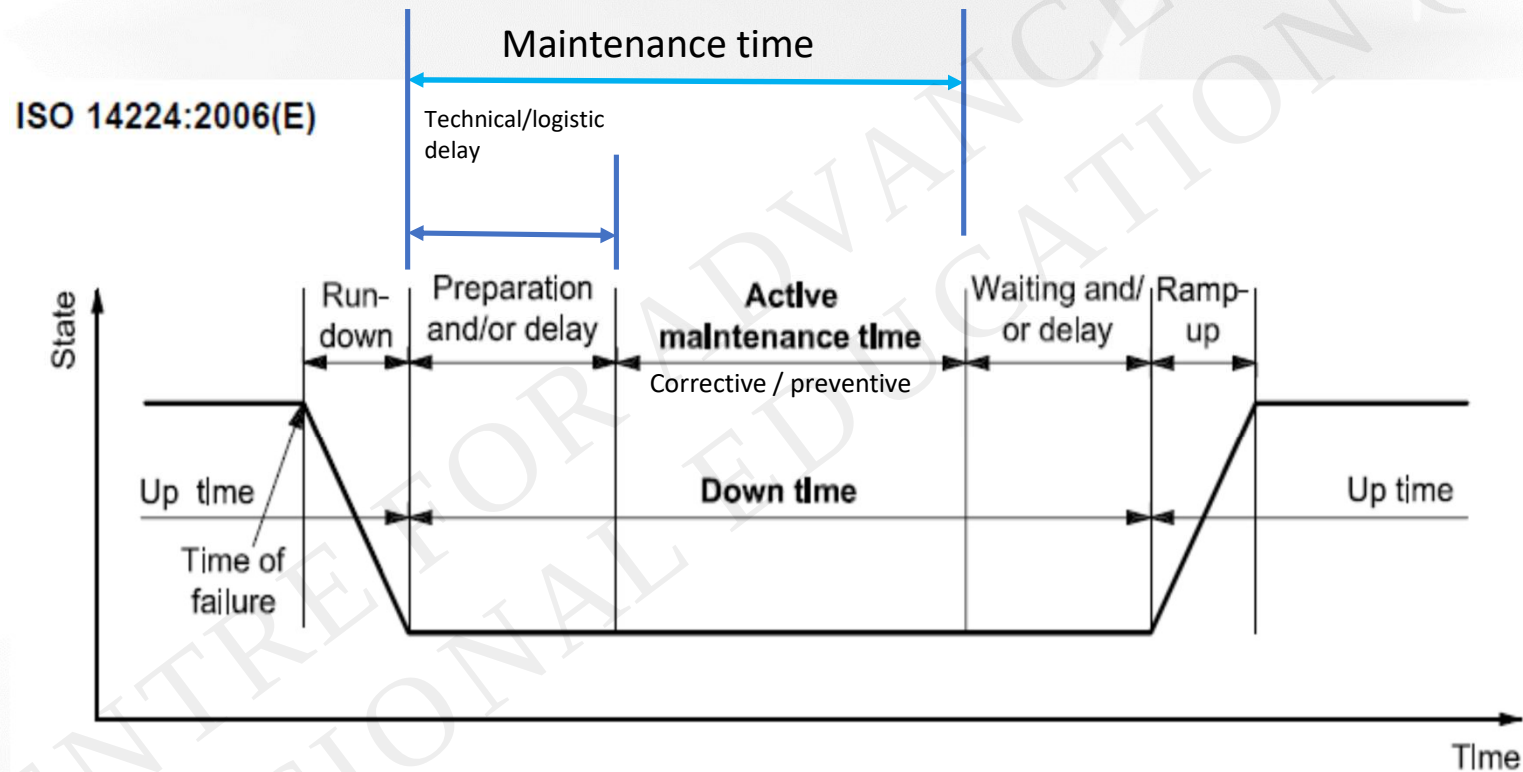


The cycle of unreliability, maintainability, availability and safety

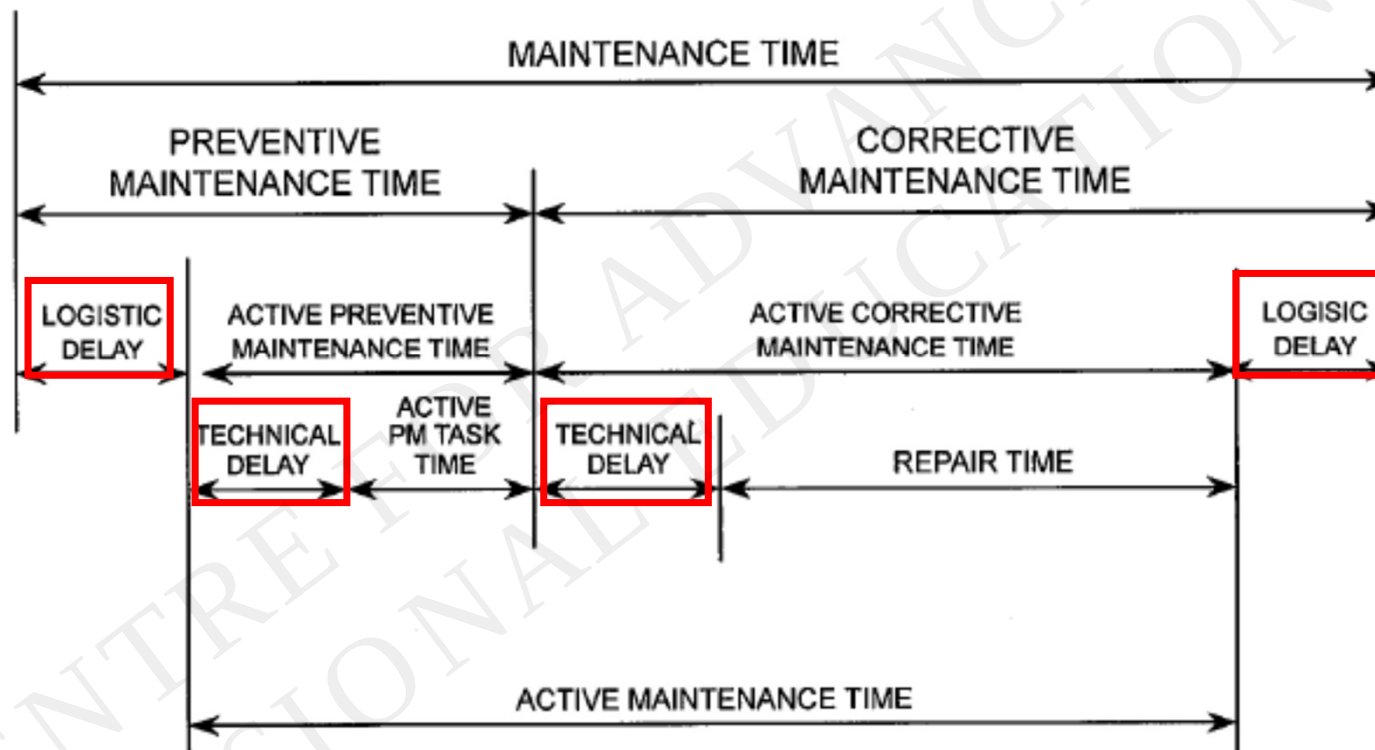


Source: Maintainability, Availability & Operational Readiness Handbook - Dr. D Kececioglu

Mean Time to Restore (MTR)



Maintenance Times



IEC 60300-3-14

Mean time to restore – *MTR* / Corrective Maintenance Time

- The downtime for MTR includes the logistic delay and administrative delay related to maintenance besides the active maintenance time. Operational delay e.g. waiting for startup are not included. Hence

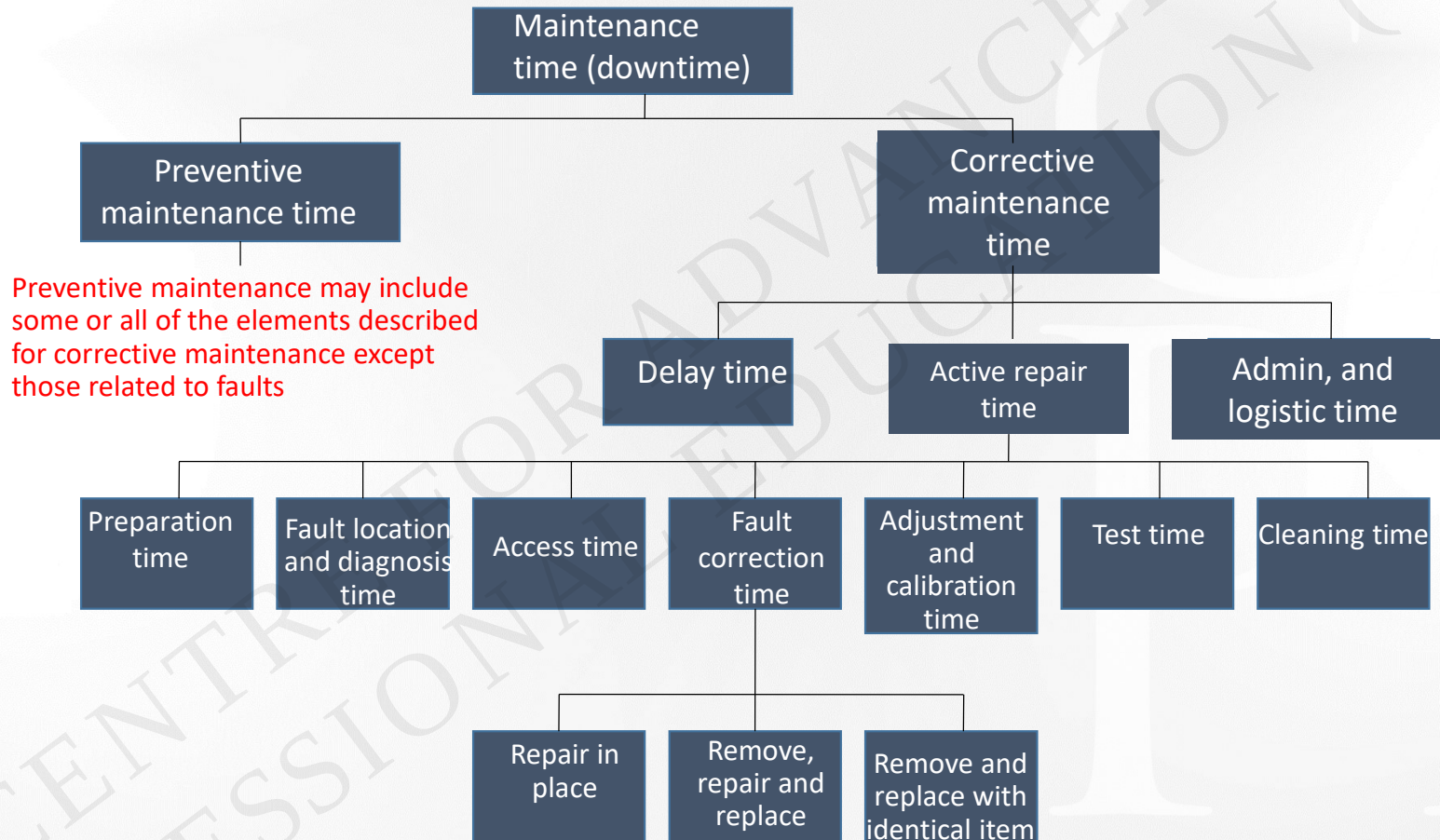
$$MTR = MTTR + LD + AD$$

where *LD* is logistic(supply) delay and *AD* is administrative delay for maintenance action. *MTTR* is as defined above.

Source: Reliability and Maintainability Eng. CE Ebeling

Equipment Downtime Components (IEC 60300-3-10)

Active and inactive downtime spent by maintainers



Delay Time

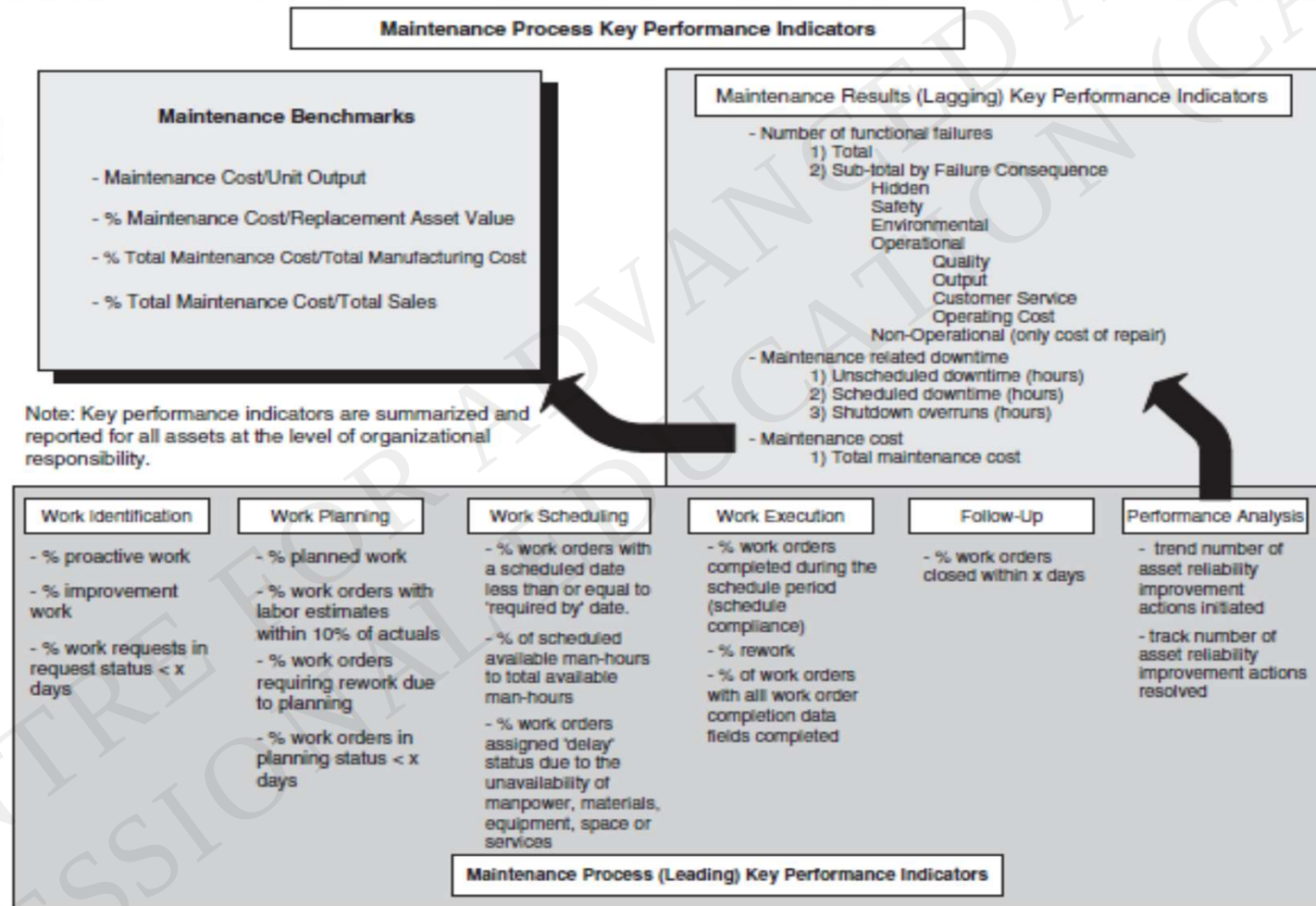
☐ Logistic Delay

- Accumulated time when maintenance cannot be carried out due to need to acquire maintenance resources, excluding any administrative delay.

☐ Technical Delay

- Accumulated time necessary to perform auxiliary technical actions associated with, but not part of, the maintenance action.

Maintenance Process Key Performance Indicators



Rules of Thumb for Maintenance and Reliability Engineers, Ricky Smith, R. Keith Mobley

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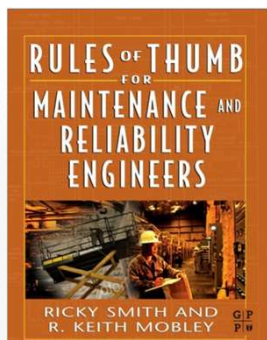
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Maintenance Support Performance Parameters/indicators – Definition

Key Performance Indicators

TABLE 6.1. Specific Examples of Leading and Lagging Indicators

KPI Type	Measure	Key Performance Indicator	World Class Target Level
Result/lagging	Cost	Maintenance cost	Context specific
Result/lagging	Cost	Maintenance cost/replacement asset value of plant and equipment	2-3%
Result/lagging	Cost	Maintenance cost/manufacturing cost	<10-15%
Result/lagging	Cost	Maintenance cost/unit output	Context specific
Result/lagging	Cost	Maintenance cost/total sales	6-8%
Result/lagging	Failures	Mean time between failure	Context specific
Result/lagging	Failures	Failure frequency	Context specific
Result/lagging	Downtime	Unscheduled maintenance related downtime (hours)	Context specific
Result/lagging	Downtime	Scheduled maintenance related downtime (hours)	Context specific
Result/lagging	Downtime	Maintenance related shutdown overrun (hours)	Context specific
Process/leading	Maintenance strategy	Percentage of work requests in "request" status for less than 5 days, over the specified time period	80% of all work requests should be processed in 5 days or less



Rules of Thumb for Maintenance and Reliability Engineers, Ricky Smith, R. Keith Mobley

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Maintenance Support Performance Parameters/indicators – Definition

TABLE 6.1. Specific Examples of Leading and Lagging Indicators

KPI Type	Measure	Key Performance Indicator	World Class Target Level
Process/leading Planning element/lagging	Planning	Percentage of work orders with work-hour estimates within 10% of actual, over the specified time period.	Accuracy of greater than 90%
Process/leading	Planning	Percentage of work orders, over the specified time period, with all planning fields completed	95% +
Process/leading	Planning	Percentage of work orders assigned "rework" status (due to a need for additional planning) over the last month.	Should not exceed 2–3%
Process/leading	Planning	Percentage of work orders in "new" or "planning" status less than 5 days, over the last month	80% of all work orders should be possible to process in 5 days or less; some work orders require more time to plan but attention must be paid to late finish date
Process/leading Scheduling element/lagging	Scheduling	Percentage of work orders, over the specified time period, having a scheduled date earlier or equal to the late finish or required by date	95%+ should be expected to ensure the majority of work orders are completed before their late finish date
Process/leading	Scheduling	Percentage of scheduled available work hours to total available over the specified time period	Target 80% of work hours applied to scheduled work
Process/leading	Scheduling	Percentage of work orders assigned "delay" status due to unavailability of personnel, equipment, space, or services over the specified time period	Number should not exceed 3–5%
Process/leading	Execution	Percentage of work orders completed during the schedule period before the late finish or required-by date	Schedule compliance of 90%+ should be achieved
Process/leading Execution element/lagging	Execution	Percentage of maintenance work orders requiring rework	Rework should be less than 3%
Process/leading	Follow up	Percentage of work orders closed within 3 days, over the specified time period	Should achieve 95%+; expectation is that work orders are reviewed and closed promptly

Rules of Thumb for Maintenance and Reliability Engineers, Ricky Smith, R. Keith Mobley

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Pareto Rules in Maintenance

- The Pareto principle (also known as the 80/20 rule, the law of the vital few, or the principle of factor sparsity) states that, for many events, roughly 80% of the effects come from 20% of the causes.
- Mid- and small-sized manufacturing companies are discovering that by following the “80/20 Rule” and instituting an efficient maintenance program, they can save an average of 30% on maintenance costs and 50% in unplanned downtime.



<https://www.eaglecmms.com/reliability-8020-formula.htm>

Pareto Rules in Maintenance

The 80/20 rule is the ideal ratio for effective equipment management:

1. Your maintenance personnel spends at least 80% of their time on preventive, proactive maintenance and 20% or less of their time in unplanned repairs.
2. No more than 20% of your equipment uses more than 80% of your maintenance time or budget. (Imagine your car, for example. While the body, interior and trunk, electrical, and cooling systems – the 80% – usually require very little maintenance or repair, the transmission and other moving parts – the 20% – require regular maintenance).

<https://www.eaglecmms.com/reliability-8020-formula.htm>

Standards in MSPA

Common standards/reference:

1. IEC 60300-3-14 – Maintenance and Maintenance Support
2. IEC 60300-3-10 - Maintainability
3. ISO 14224 - Collection and Exchange of Reliability and Maintenance Data for Equipment
4. IEC 60706-4 - Maintenance and Maintenance Support Planning
5. IEC 61703:2001 – Mathematical Expressions for Reliability, Availability, Maintainability and Maintenance Support Terms
6. IEC 60300-3-12 – Integrated Logistic Support

Maintenance Support Objectives

Improvement in maintenance and maintenance support can be achieved by changes in:

➤ **Maintenance**

- Maintenance procedures
- Equipment and system design
- Maintainability of an item

➤ **Maintenance Support**

- Coordination of operation
- Spare parts and materials
- Tools and support equipment
- Use of external resources

RAM and MSPA

RAM Methodology

Input

Eqp. Running Hrs Record

Equipment	Running Hrs	Failure Mode	Time to Repair	Cost
1	1000	Failure Mode 1	100	1000
2	2000	Failure Mode 2	200	2000
3	3000	Failure Mode 3	300	3000
4	4000	Failure Mode 4	400	4000
5	5000	Failure Mode 5	500	5000
6	6000	Failure Mode 6	600	6000
7	7000	Failure Mode 7	700	7000
8	8000	Failure Mode 8	800	8000
9	9000	Failure Mode 9	900	9000
10	10000	Failure Mode 10	1000	10000

Display Hist. Proactive Rectification: Central Header
Person responsible: [Name]
Equipment: [ID]
Location: [Location]

- PMMS:**
- Man hours
 - Spare Parts
 - Damage code
 - Cost

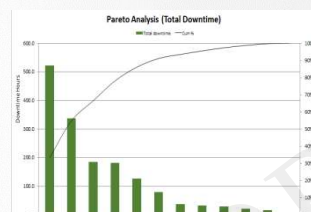
Operation	Start Date	End Date	Act. Start	Act. End
1	10/10/2014	10/10/2014	10/10/2014	10/10/2014
2	10/10/2014	10/10/2014	10/10/2014	10/10/2014
3	10/10/2014	10/10/2014	10/10/2014	10/10/2014
4	10/10/2014	10/10/2014	10/10/2014	10/10/2014
5	10/10/2014	10/10/2014	10/10/2014	10/10/2014
6	10/10/2014	10/10/2014	10/10/2014	10/10/2014
7	10/10/2014	10/10/2014	10/10/2014	10/10/2014
8	10/10/2014	10/10/2014	10/10/2014	10/10/2014
9	10/10/2014	10/10/2014	10/10/2014	10/10/2014
10	10/10/2014	10/10/2014	10/10/2014	10/10/2014

EFR

Failure Mode	Failure Description	Failure Date	Failure Time	Failure Location
1	Failure Mode 1	10/10/2014	10/10/2014	10/10/2014
2	Failure Mode 2	10/10/2014	10/10/2014	10/10/2014
3	Failure Mode 3	10/10/2014	10/10/2014	10/10/2014
4	Failure Mode 4	10/10/2014	10/10/2014	10/10/2014
5	Failure Mode 5	10/10/2014	10/10/2014	10/10/2014
6	Failure Mode 6	10/10/2014	10/10/2014	10/10/2014
7	Failure Mode 7	10/10/2014	10/10/2014	10/10/2014
8	Failure Mode 8	10/10/2014	10/10/2014	10/10/2014
9	Failure Mode 9	10/10/2014	10/10/2014	10/10/2014
10	Failure Mode 10	10/10/2014	10/10/2014	10/10/2014

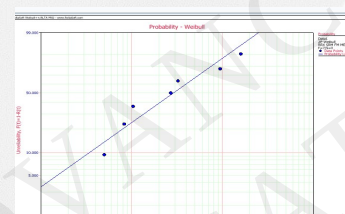
Exploratory Analysis

- Identify Failure Mode
- Calculate Time Between Failure
- Calculate Time To Repair

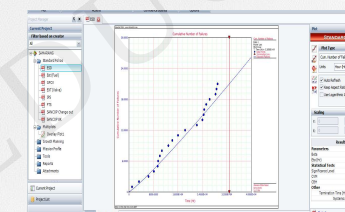


- Pareto Analysis:**
- Failure Modes
 - Failure Downtime

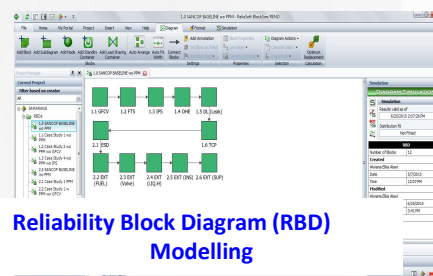
Modelling



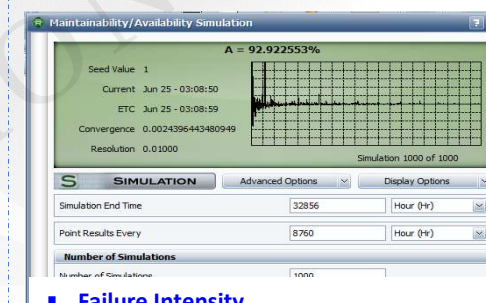
Weibull++ - Identify Failure Distribution, MTBF and MTTR



Reliability Growth Analysis (RGA) Modelling

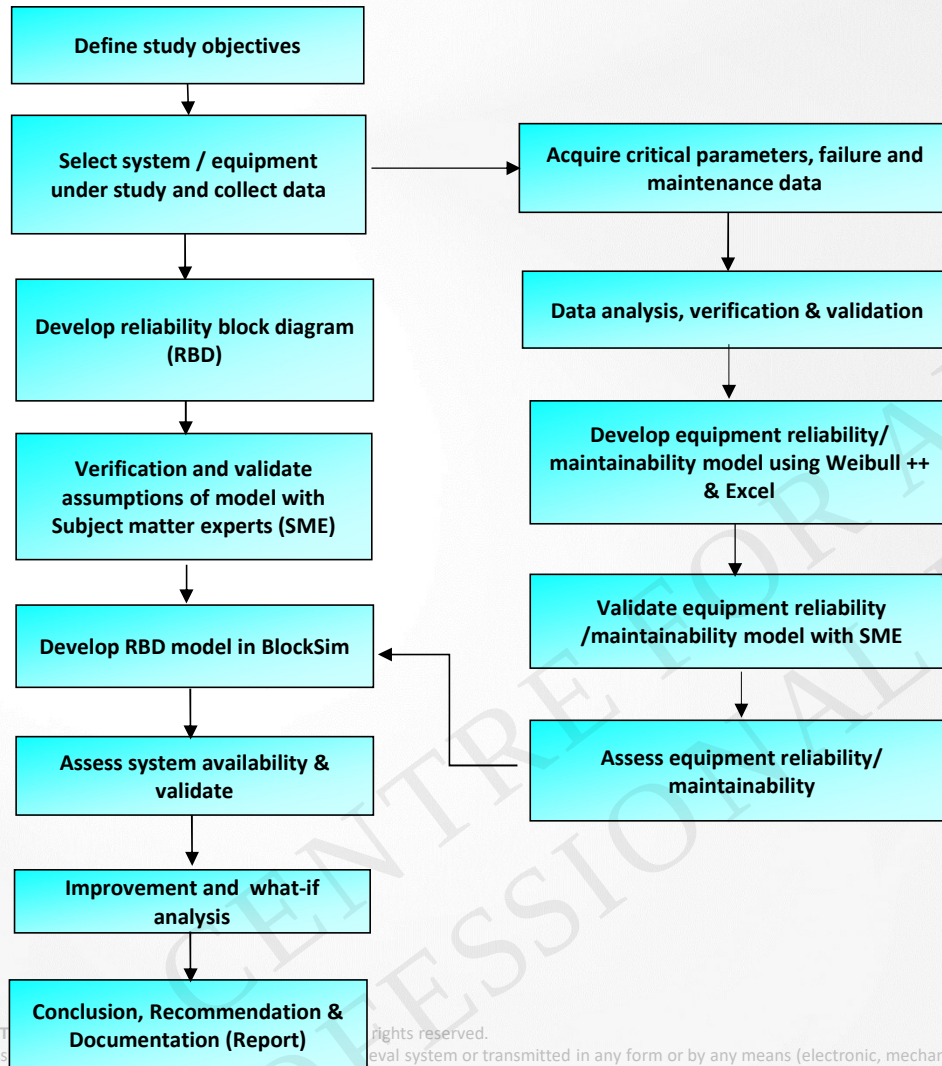


Results

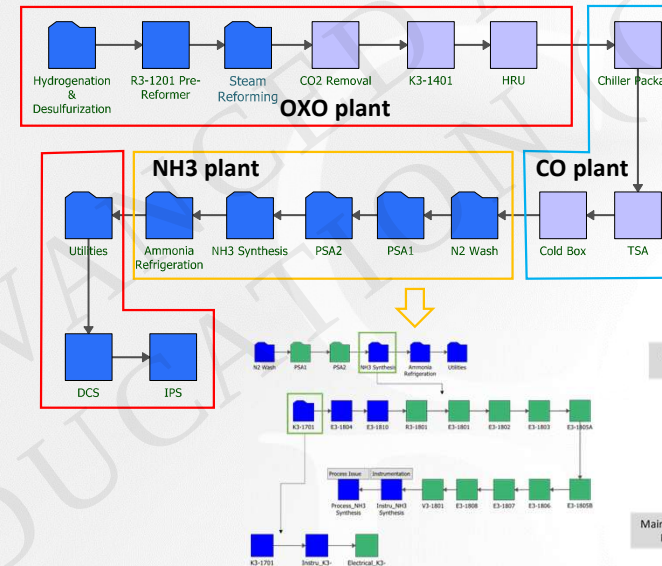


- Failure Intensity
- Equipment Availability
- With and Without PPM (Current, Forecast, Failure Frequency Prediction)
- Sensitivity Cases

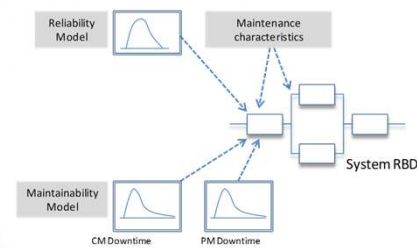
RAM Methodology



Example



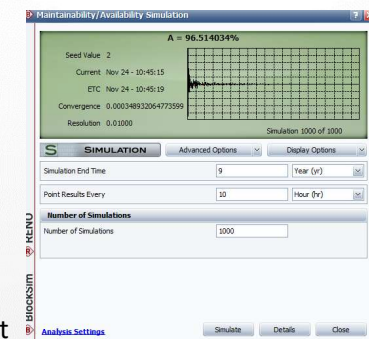
RBD



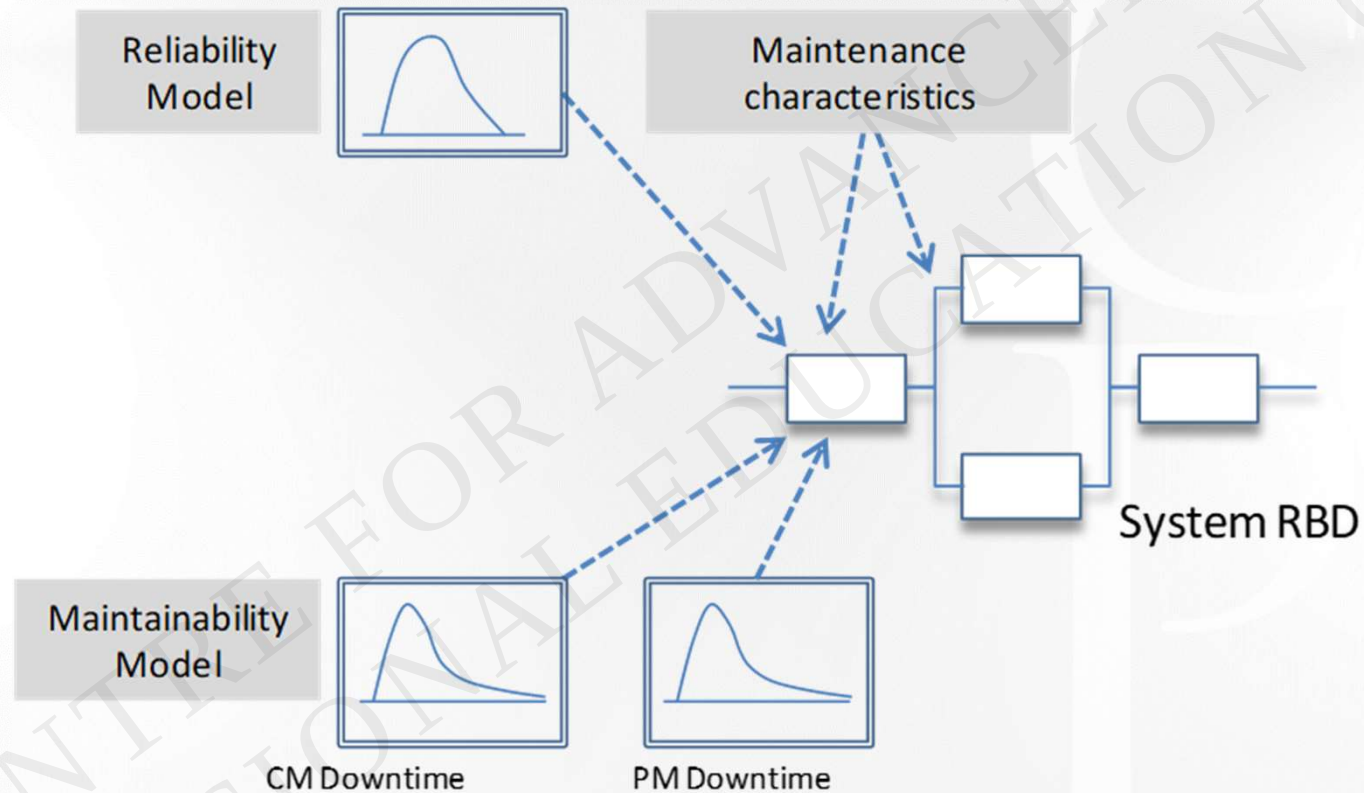
Model Validation

Plant	Simulation PR	Plant PR	Abs Error	PR target
OXO	96.51	95.89	0.62	98.1
NH3	98.35	98.32	0.03	97.6
CO	98.91	99.56	0.65	97.6

OXO Plant simulation results



Reliability Block Diagram (RBD)



Availability

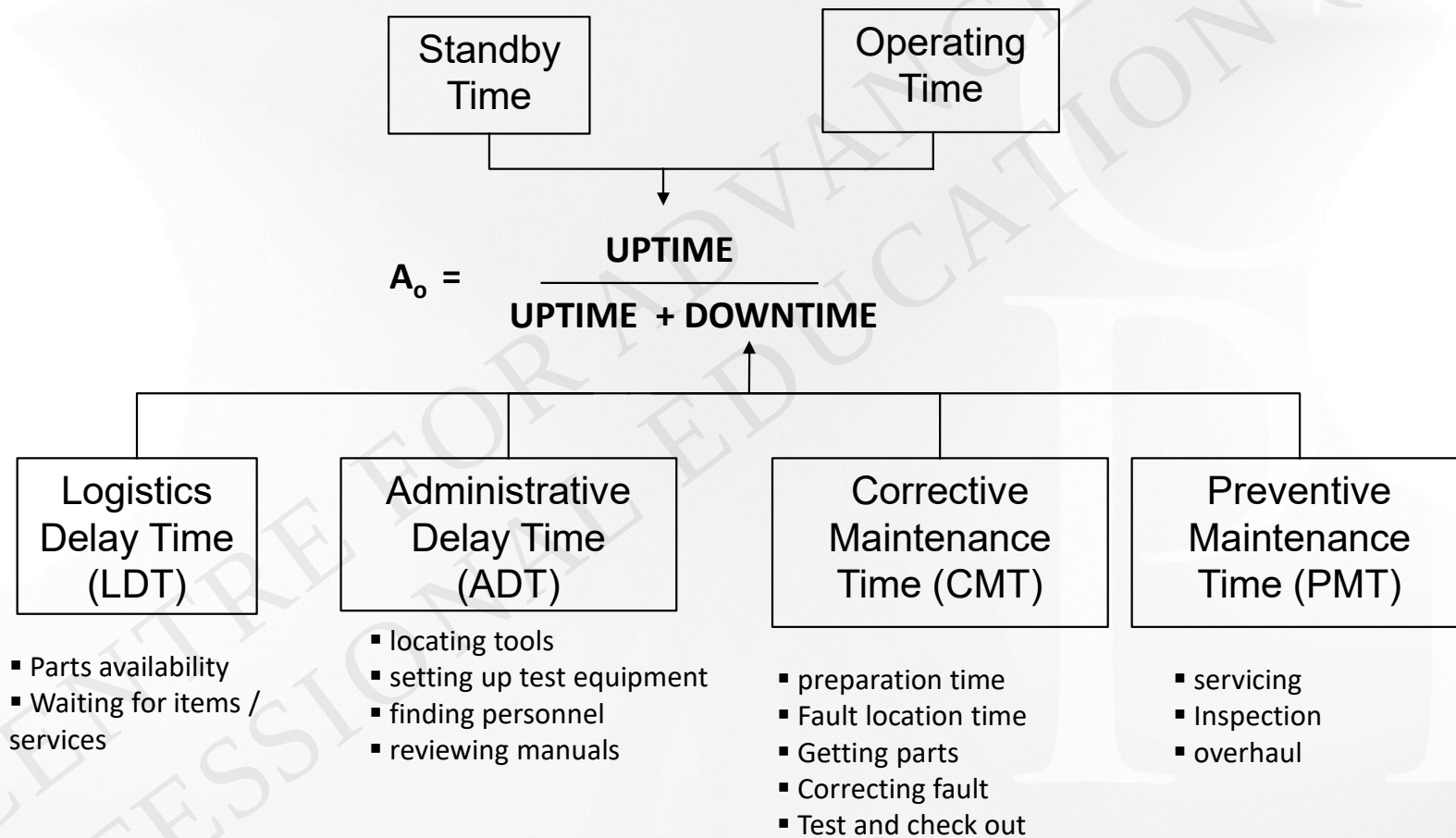
Definition

“The probability that a system or component is performing its required function at a given point in time or over a stated period of time when operated and maintained in prescribed manner”

(Ebeling, 1997)

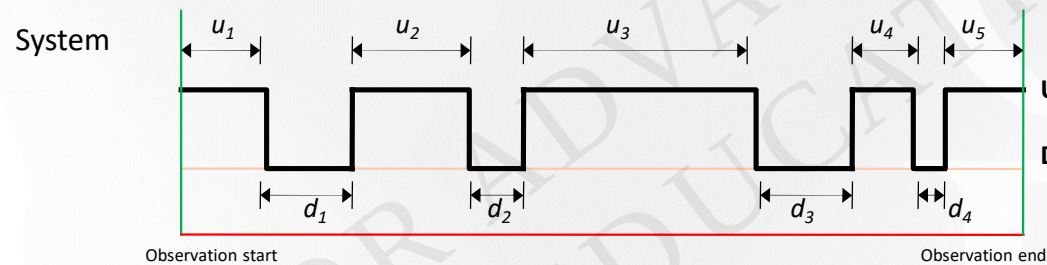


Operational Availability



System Availability

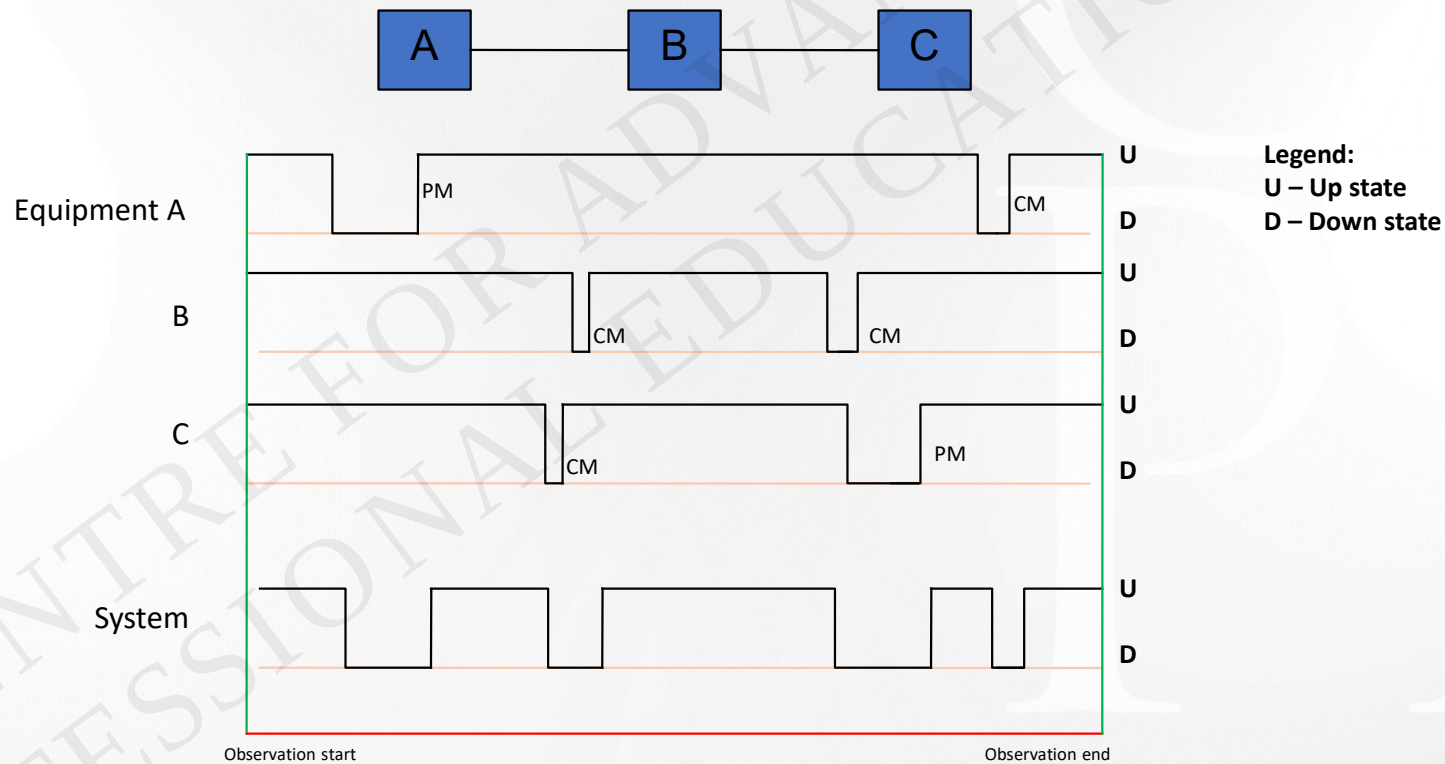
$$\text{System Availability} = \frac{\text{UPTIME}}{\text{UPTIME} + \text{DOWNTIME}}$$



$$\text{System Availability} = \frac{\sum_{i=1}^n u_i}{\sum_{i=1}^n u_i + \sum_{i=1}^n d_i}$$

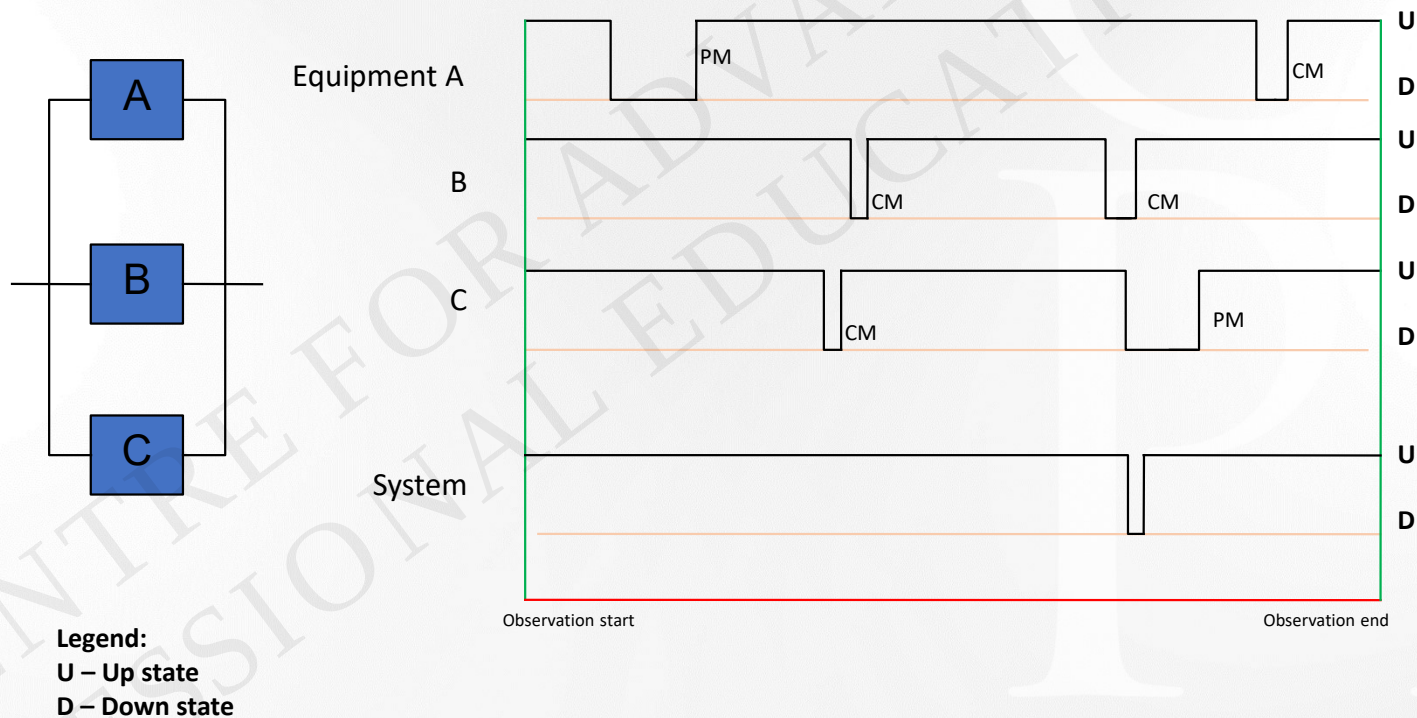
System Availability

Consider a system having 3 equipment in series. A down state of any equipment will bring the system down. Equipment can be in down state due to CM and PM actions



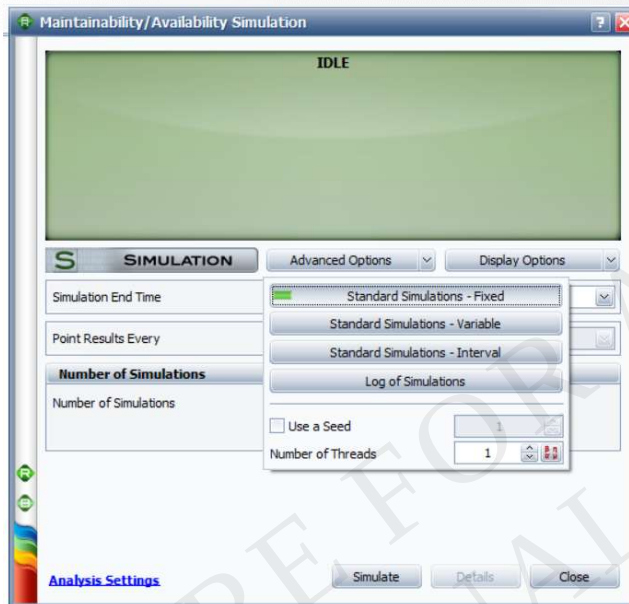
System Availability

Now consider a system having 3 equipment in parallel configuration. The system is down when 2 equipment are in down state (2-out-of-3)

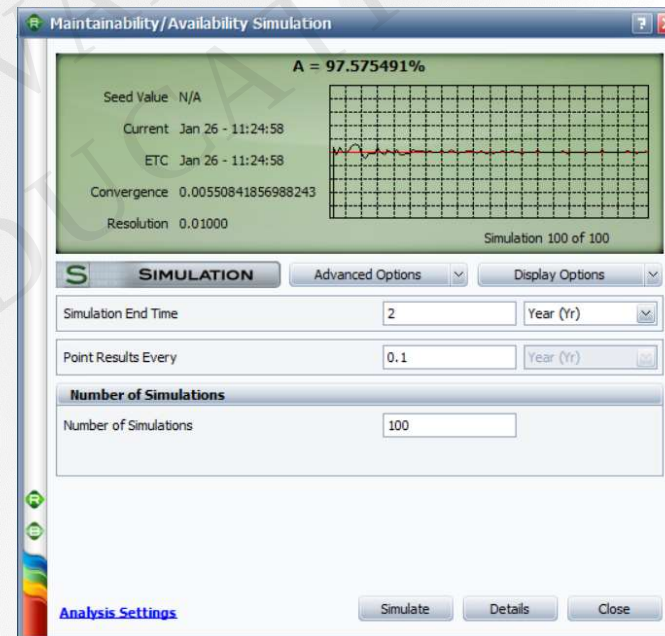


Availability Calculation in BlockSim

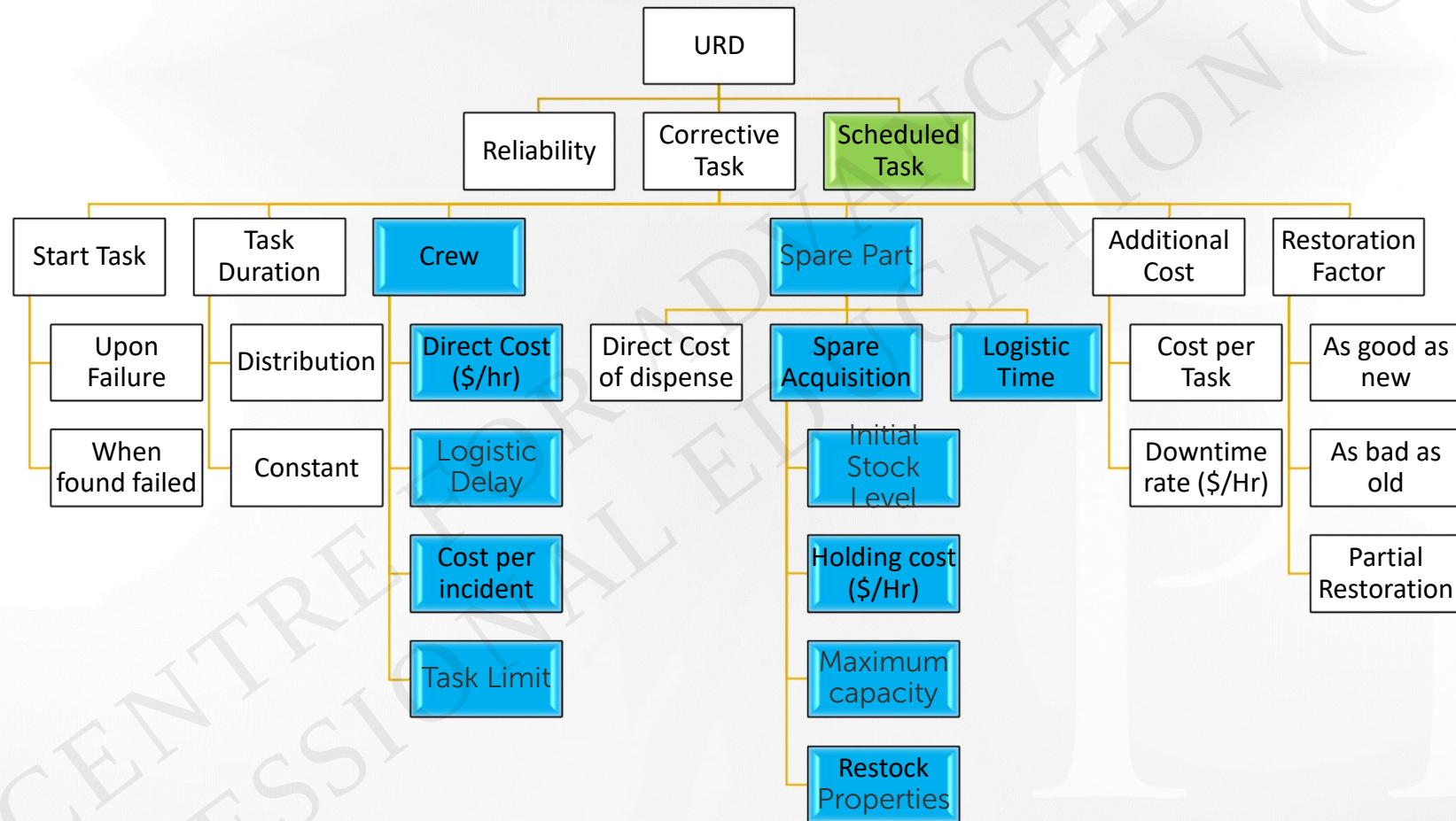
Simulation Setup



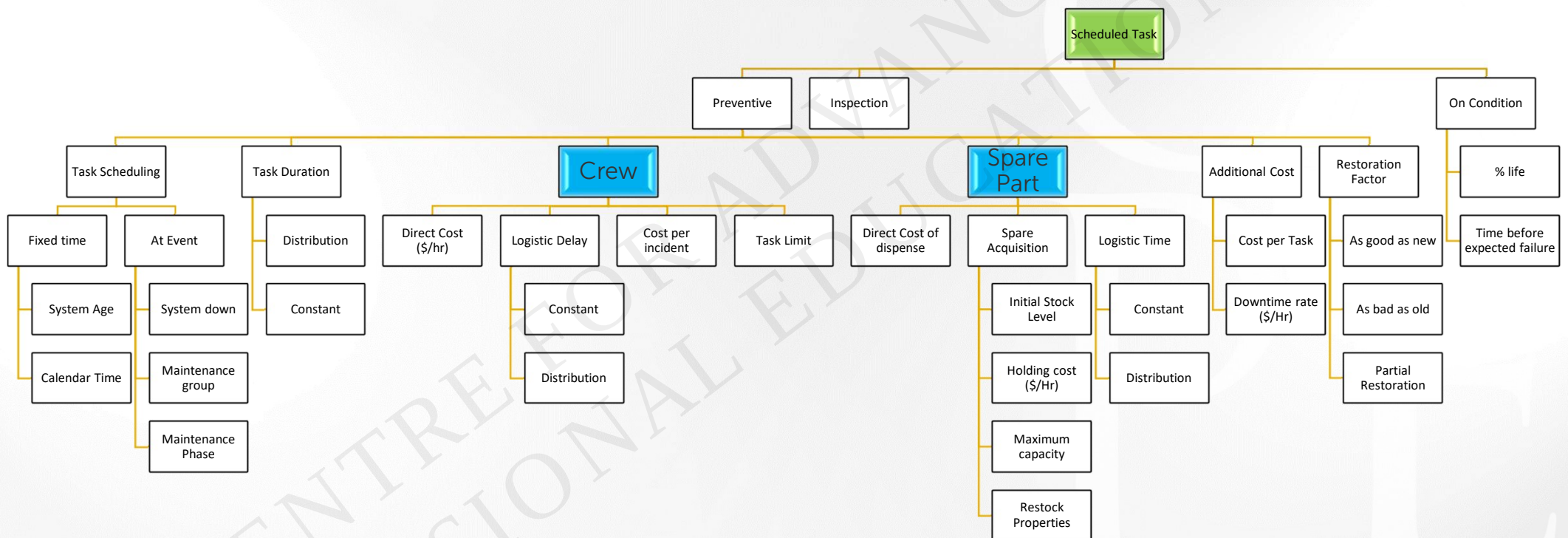
Simulation Result



BlockSim Simulation: Maintenance Support



Maintenance Support for Scheduled Task



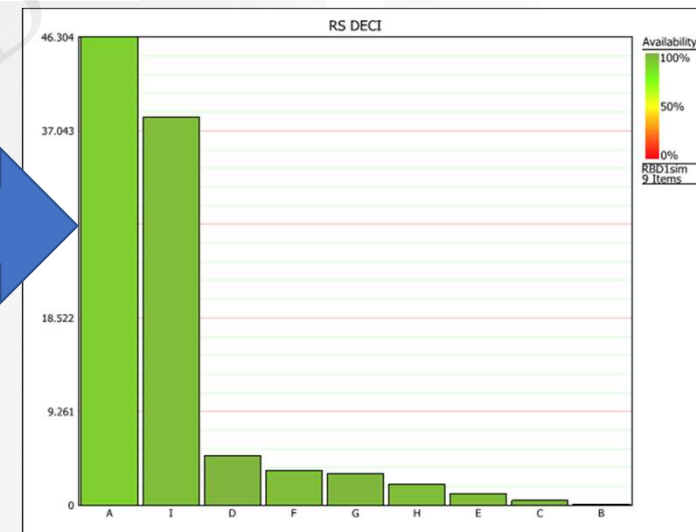
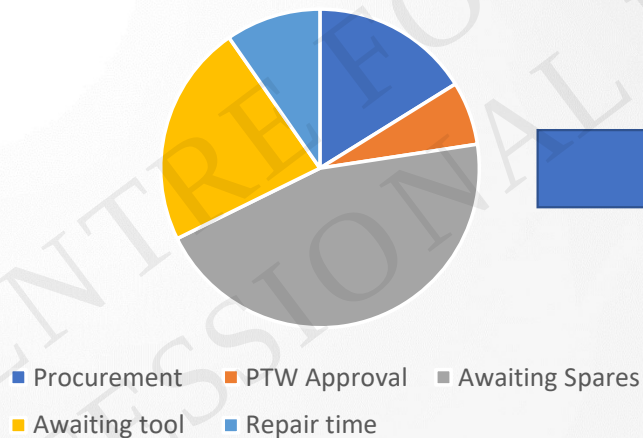
Identifying Critical component/System

- **RS DECI (Downing Event Criticality Index)**
 - The index is a relative measure of the number of system downing events caused by this component.
 - Percentage of times that any downing event of this component caused the system to go down.

$$RS\ DECI = \frac{\text{Number of System Downing Events (Component)}}{\text{Number of System Downing Events}}$$

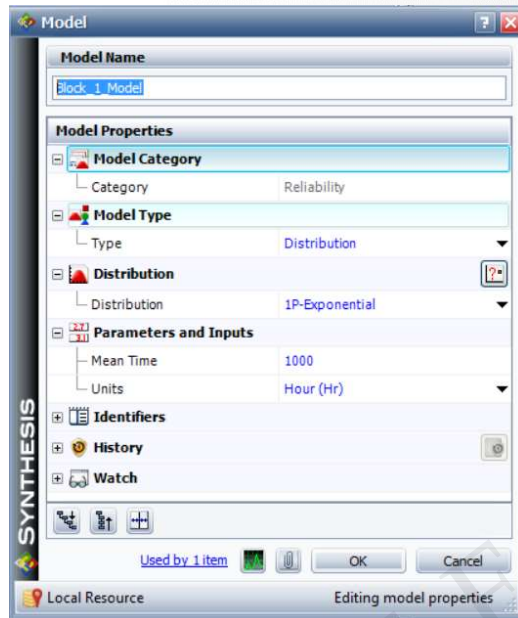
- It includes all downing events.

Maintenance Activities

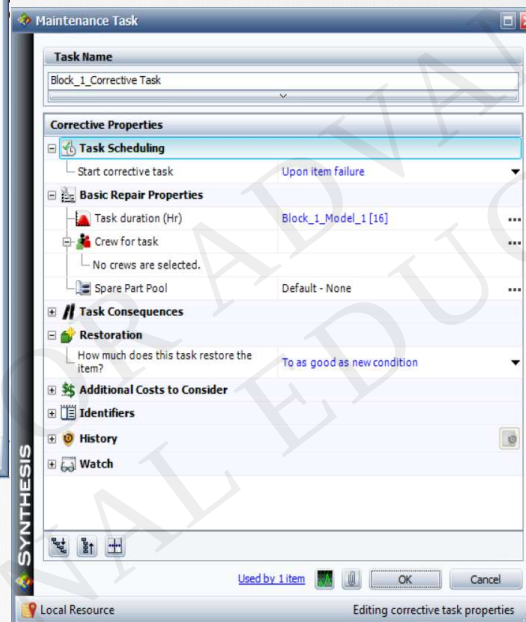


Availability Calculation in BlockSim

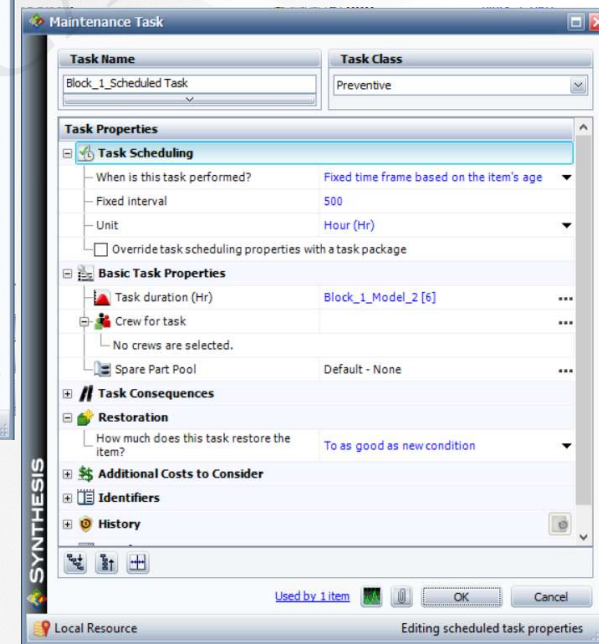
Reliability



Repair



PM



Data Collection in Maintenance

Data collection for Maintenance Support Performance Parameter/Indicators

- It is necessary to precisely define
 - ☐ The data to be collected in order to determine the values required for the indicator
 - ☐ The measurement method (Operating mode)
 - ☐ The tools required for the measurement (documents, counters, sensors, analyzers, computerized maintenance management system, etc.)
- To make the possible evaluation and comparisons easier, it is obligatory that the collected data are in conformity with the standardized definitions when they exist (see EN 13306). If the definition does not exist, it will be necessary to derive such a definition.

Analysis method and data requirements

What is it for?	Data requirement	IEC reference	Title
How to achieve and verify maintenance objectives	Reason for maintenance actions, type of maintenance action completed, man-hours for maintenance action and elapsed calendar time for same. Total downtime, operating hours, number and skill level of maintenance team, utilization of test and check-out equipment and spares consumption	IEC 60706-3	Verification and collection, analysis and presentation of data
How to use statistical methods in maintenance evaluation	Times required to perform maintenance on specified equipment	IEC 60706-6	Statistical methods in maintainability evaluation

Maintenance Data Collection

- Maintenance data collected during the operational use of an item represent the optimum source of information on the quantitative maintainability characteristic.
- To enable proper conclusions to be drawn from these data, they should include, as a minimum
 - ☐ Identification of the item, assembly, subassembly, etc.
 - ☐ Reason for the maintenance action
 - ☐ Type of maintenance action performed, this may be either scheduled maintenance e.g. servicing, inspection, calibration, overhaul, renewal of parts) or unscheduled maintenance (e.g. repair on site, removal and replacement, shop repair)
 - ☐ Man-hours for maintenance action (Crew Time)
 - ☐ Elapsed time for maintenance action (Calendar Time)
 - ☐ Total maintenance downtime (active maintenance downtime, maintenance delay time)

Data Acquisition and Analysis

Adjustments or censoring of the data may be necessary for one or more of the following reasons:

1. System similarity
 - Is the item that provided historical data similar to the one to be evaluated
2. Design modifications
 - Has the validity of the data been affected by any design configurations changes such as access?
3. Maintenance concept
 - Is data transferability influenced by different maintenance concept?
4. Maintenance conditions
 - Were the data based upon the same standards, such as personnel skill, maintenance tools and facilities, check out equipment

Required Data

1. Failure data

- Failure data for characterizing a failure, e.g. failure data, items failed, failure impact, failure mode, failure cause, failure detection method

2. Maintenance data

- Maintenance data, parameters characterizing a maintenance action, e.g. date of maintenance , maintenance category, maintenance activity, impact of maintenance, items maintained.
- Maintenance resources, maintenance man hours per discipline and total, utility equipment/resources applied
- Maintenance times, active maintenance time, down time

Time Study

What is Time Study?

- Time study is a work measurement technique for **recording the time of performing a certain specific job** or its element carried out under specific condition and for analysing the data so as to obtain the time necessary for an operator to carry out at a defined rate of performance.
- Time study is a method of direct observation.
- A trained observer watches the job and records data as the job is being performed over a number of cycles.

Time Study Equipment

Time study equipment



Decimal
minutes

Decimal
hours

Selection of the job

- The job in question is a new one, not previously carried out (new product, component, operation or set of activities).
- A change in material or method of working has been made and a new time standard is required.
- A complaint has been received from a worker or worker's representative about the time standard for an operation.
- A particular operation appears to be a “bottleneck” holding up subsequent operations and possibly previous operations.

General Rules

- Before starting the study, there are a number of points to be observed in relation to the approach to the worker and the job he is doing.
- Make sure the job has been method studied. Especially where the standard time is to be used for incentive purposes.
- In an organization where time study has never been used before, supervision and worker representatives should be told the reasons for the study program and given and insight into the study procedure.
- It is good policy to ask the supervisor or workers representatives to be studied. The worker should be a competent steady person with an average rate of working.

General Rules

- The worker should be told the reasons for the study. Why he has been selected and asked to work at his normal pace explaining any problems that may arise during the time the study is taken.
- Workers should be allowed plenty of time to settle down to new methods.
- When making the study the observer should take a position where he can see the whole of the job being performed. This is generally to the side of the operator, slightly to the rear and approximately 6 feet away. No attempt should be made to carry out timing from a concealed position, without the operators knowledge or with the watch in the pocket.

Element

- An element is a distinct part of a specified job selected for convenient of observation - measurement and analysis.
- How to breakdown the job in to elements?
 1. Elements should be easily identified, with definite beginnings and endings so that, once established. They can be repeatedly recognized. The point at which one element ends and another begins is called break point. Breakpoints must be decided by the work study officer when he divides the work cycle into elements.
 2. All breakpoints can be recognized by a sound or by a change of direction of a hand or arm.
 3. Elements should be as short as can be conveniently timed by a trained observer, but should not be less than 0.10 minutes and more than 0.50 minutes.
 4. Manual elements should be separated from machine elements for future use as synthetic data.
 5. Short elements should be next to long elements.

Reasons for breaking down the job into elements

- Variations in speed, pace and effort which an operator works can be more accurately recorded over short periods of time. There could be some parts of the work-cycle with the operator performs better than others.
- The more clear-cut the item of work which is timed, separately, the more readily can an appropriate fatigue allowance be arrived at for incorporation with the element.
- To ensure that productive work is separated from unproductive work.
- To enable a detailed job specification to be produced.

Standard time

- Standard time = time that is allowed to perform the job satisfactory.



Types of Allowances

1. Relaxation Allowances

- Relaxation allowance is an addition to the basic time intended to provide the worker with the opportunity to recover from the physiological and psychological effects of carrying out specified work under specified conditions and to allow attention to personal needs. The amount of allowance will depend on the nature of the job.
- There are 2 type of relaxation allowance
 1. Personal needs
 - This allowance provides for the necessity to leave the workplace to attend to personal needs such as washing, going to the lavatory or fetching a drink
 - Common figures are from 5 to 7 percent of basic time.
 2. Basic fatigue
 - This allowance, always a constant is given to take account of the energy expended while carrying out work and to alleviate monotony.
 - A common figure is 4 percent of basic time.

2. Contingency allowances

- A contingency allowance is a small allowance of time which may be included in a standard time to meet legitimate and expected items of work or delays, the precise measurement of which is uneconomical because of their infrequent or irregular occurrence.

3. Various other types of allowance are

- Policy allowance
- Special allowance

Steps in making a time study

1. Obtaining and recording all the information available about the job, the operative and the surrounding conditions , which is likely to affects the carrying out the work.
2. Recording a complete description of the method, breaking down the operation into elements.
3. Examination the detailed breakdown to ensure that the most effective method and motions are being used, and determining sample size.
4. Measuring with timing device and recording the time taken by the operative to perform each element of operation
5. At the same time, assessing the effective speed of working of the operative relative to the observer's concept of the rate corresponding to standard rating.
6. Extending the observed time to basic time.
7. Determining the allowances to be made over and above the basic time for the operation.
8. Determining the standard time for the operation.

Work Sampling Study

Work Sampling Study

- A method used to estimate to a high degree of accuracy the population parameters such as mean and standard deviation of random variables such as proportion of “success” in the population.
- It is a rather inexpensive method.
- It only requires small samples to be taken out of the whole population.
- It is somewhat different from time-motion studies.

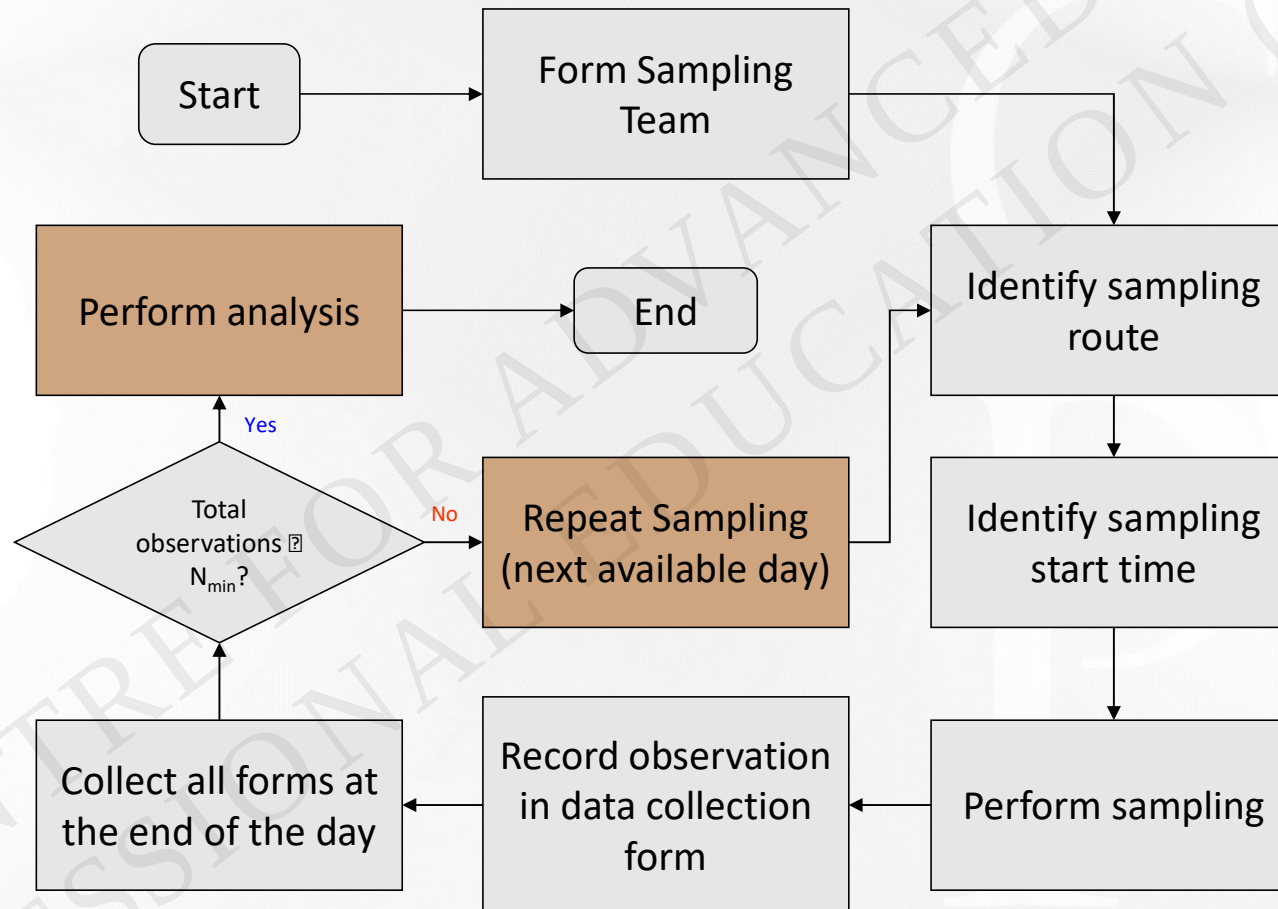
Work Sampling Method: How does it suit our purposes?

- Random samples are required.
- As the samples are taken randomly, this would avoid biasness in the measurement of the population parameters, i.e. proportion of “success”.
- Each observation data for work sampling is a snapshot of activities and is recorded into different distinct categories.
- Sampling is repeated to increase accuracy

Conduct of Work Sampling Study

- Conduct openly with full knowledge of employees, i.e. announcement before study is made.
- No name, place or time of study is indicated (fully random).
- Objective is to measure effectiveness of manpower procedure, process and supervision; not individuals.

Work Sampling Process



How to record data

- Identify activity observed and assign the appropriate activity code
- Record the number of observation made for each of the identified activity
- Make sure that the data collected are as representative as possible and avoid any biased observation

[Click here for
Activity Codes](#)

[Click here for
Observation Form](#)

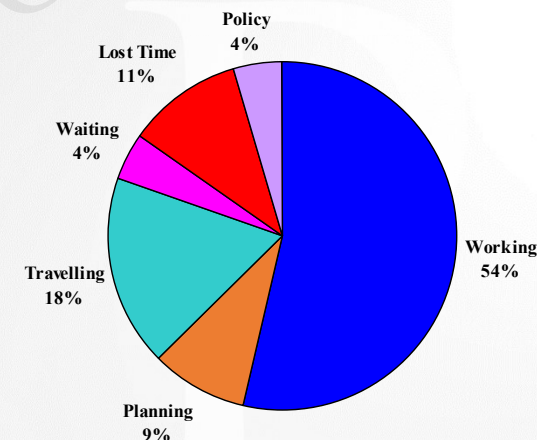
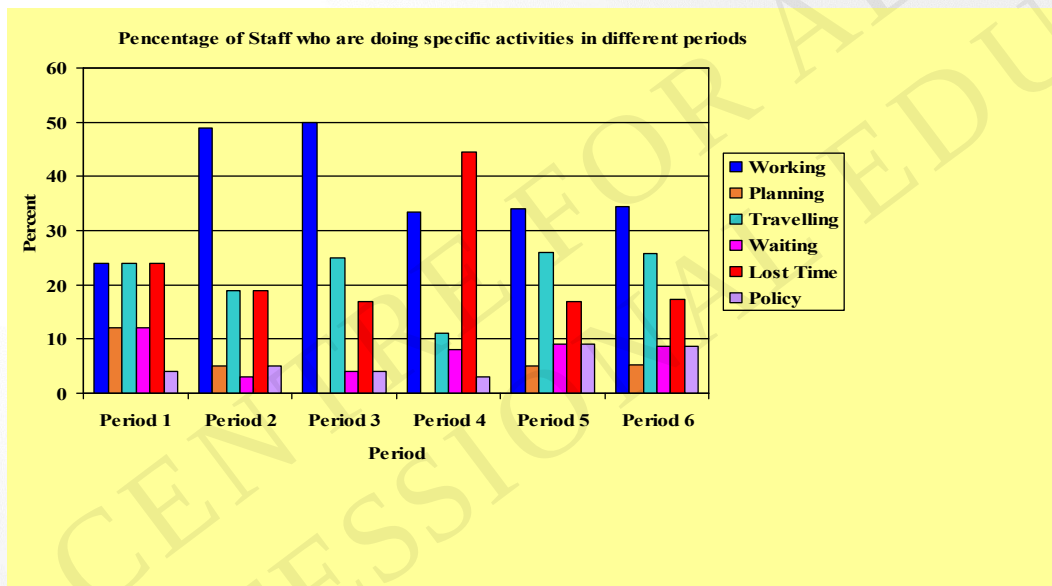
How accuracy is obtained

- Collect the number of observations to the required minimum for the accuracy required
- Results are based on total observations recorded and within the confidence level and standard error specified (95% confidence & $\pm 2.5\%$ error)
- If the result of observation is far away from estimate of 0.5 used to determine the total number of required observations, and if the number of observations obtained is equal to or greater than the required number, the standard error for this observation will be smaller, i.e. more accurate result is obtained.

[Click here for Error Table](#)

Presentation of Results

- Data and results can be presented in forms of diagrams and charts, e.g. bar chart, trend chart or pie diagram
- The results can then be compared with benchmark to determine the next appropriate action plans for improvement and any recommendations



Assumptions

- Each period has its own distribution which are repeated in every normal working day
- Each sampling day represents actual condition during normal working days

Work Sampling Activity Codes

Working

Tools in Hand	01
Layout/Measurement	02
Helping	03
Housekeeping	04
Safety Precaution	05
Receiving Instructions	06
Testing/Inspection	07

Planning/Leading

Coordinating/Planning	11
Paperwork	12
Observing/Sampling	13
Checking Drawings	20
Checking Instructions	21
Checking Schedule	22
Checking Safety Precautions	23
Checking Materials	24
Checking Tools	25
Checking Equipment	26
Checking Mobile Units	27

Waiting

For Instructions	30
For Operations	31
For Mobile Unit	32
For Safety Inspection	33
For Safety Permit	34
For Safety Equipment	35
For Group Member	36
For Other Craft Group	37
For Tools	38
For Materials	39
For Transportation	40

Travelling

Between Jobs	50
For Materials/Area	51
For Materials/Warehouse	52
For Tools/Area	53
For Tools/Outside Area	54
For Tools/Tool room	55
To/From Rest	56
To/From Lunch	57
To/From Showers	58
Travel Within Job	59

Lost Time

Unassigned	60
Personal	61
Late Start	64
Early Quit	65
Idle in Field	66
Idle in Shop	67

Plant Policy

Training Class	90
Medical/Physical Examination	91
Medical/First Aid	92
Lunch	93
Smoke/Coke/Rest	94
Showers/Locker	95
Clocking In/Out	96
Safety Meetings	97
Information Meetings	98
Travel to/from Safety Meeting	99



Observation form

TIME STARTED _____

TIME FINISHED _____

DATE _____

WEATHER _____

OBSERVER _____

PLANT _____

AREA _____

ELEMENTS		OBSERVATIONS	TOTAL
WORK	Working (please specify code)		
	Assisting 03		
	Planning/Leading (please specify code)		
	Other (please specify code)		
	TOTAL WORK OBSERVATIONS		A
TRAVEL	To and From Job 50/59		
	To Get Tools 53/54/55		
	To Get Materials 51/52		
	Other (please specify code)		
	TOTAL TRAVEL OBSERVATIONS		
WAIT	Tools 38		
	Material 39		
	Process 31		
	Safety 33/34/35		
	Instructions 30		
DELAYS	Other Craft 37		
	Other (please specify code)		
	Craft Delays		
	Wrong Tools		
	Union Meeting		
MISCELLANEOUS	Safety Meeting		
	Other (please specify code)		
	Late Start 64		
	Early Quit 65		
	Idle 66/67		
LOST TIME	Personal 61		
	Other (please specify code)		
	TOTAL LOST TIME OBSERVATIONS		B
TOTAL DELAY AND WORK OBSERVATIONS		A+B	

PERCENT

WORKING

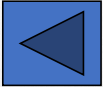
DELAY

%

%

SHEET

of

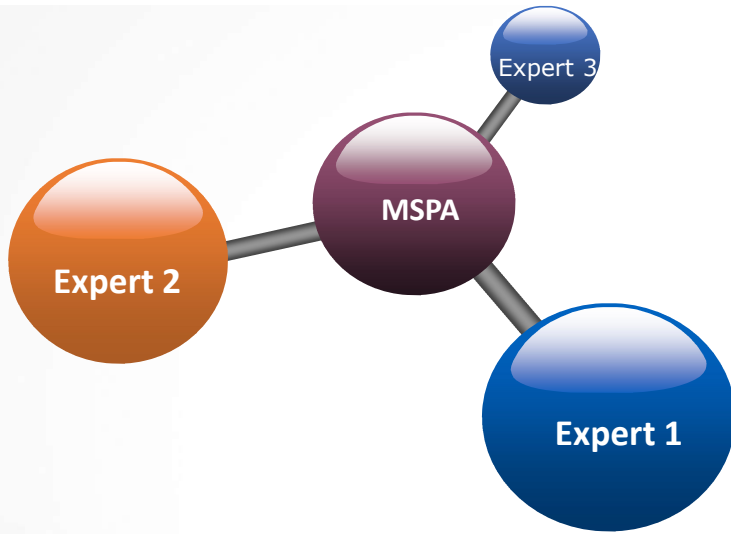


Sampling accuracy/error distribution table

<i>p</i>	<i>e</i>				
	3.0%	2.5%	2.0%	1.5%	1.0%
0%	0	0	0	0	0
1%	43	61	96	170	381
3%	125	179	280	497	1,118
5%	203	292	457	812	1,825
10%	385	554	865	1,537	3,458
15%	545	784	1,225	2,177	4,899
20%	683	984	1,537	2,732	6,147
25%	801	1,153	1,801	3,202	7,203
30%	897	1,291	2,017	3,586	8,068
35%	972	1,399	2,185	3,885	8,740
40%	1,025	1,476	2,305	4,098	9,220
45%	1,057	1,522	2,377	4,226	9,508
50%	1,068	1,537	2,401	4,269	9,604
55%	1,057	1,522	2,377	4,226	9,508
60%	1,025	1,476	2,305	4,098	9,220
65%	972	1,399	2,185	3,885	8,740
70%	897	1,291	2,017	3,586	8,068
75%	801	1,153	1,801	3,202	7,203
80%	683	984	1,537	2,732	6,147
85%	545	784	1,225	2,177	4,899
90%	385	554	865	1,537	3,458
95%	203	292	457	812	1,825
98%	84	121	189	335	753
99%	43	61	96	170	381
100%	0	0	0	0	0

At 95% symmetrical
confidence interval
($z = 1.96$)





Elicitation of Expert Opinion

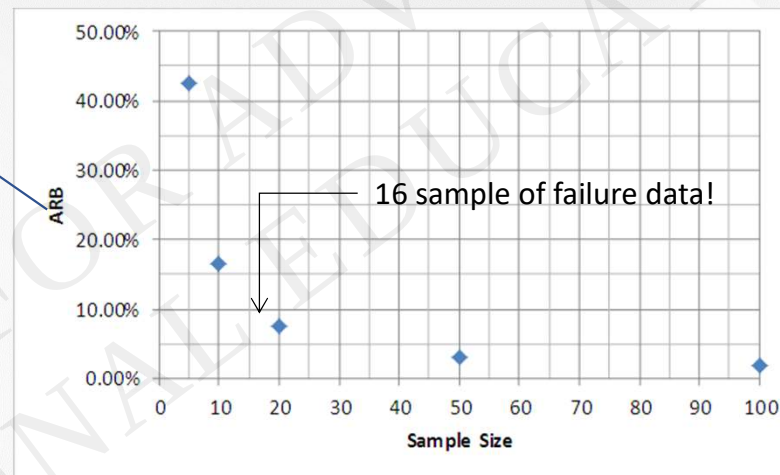
Background

- Acceptable estimation for MSP depends on availability of reliable data.
- MSP data scarcity due to several factors:
 - Data is never captured properly
 - Data is not captured properly - Data is available but not reliable
 - Changes to equipment design or maintenance causing historical data irrelevant
- Expert opinions can be utilized as another valuable data source

Effect of sample size to relative bias

- Relative bias is the difference between the estimated parameters to the true values
- Monte Carlo simulation result show that to get the bias of less than 10% requires at least sample size 16

Case study based
on Weibull
Distribution



Normally we
don't have this
many data nor
we want to have
this many
failures!

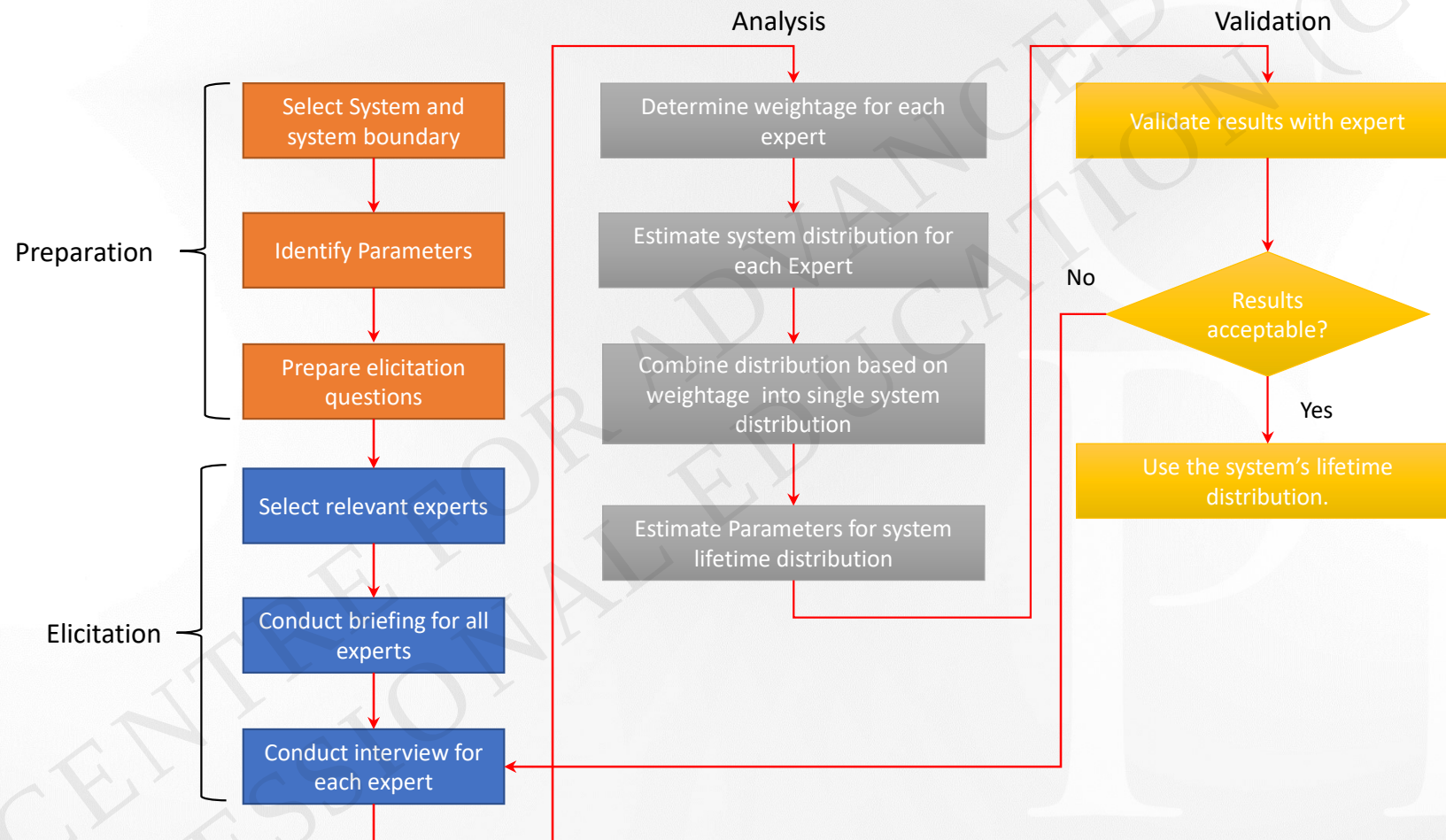
$$RB = \frac{(\hat{\beta} - \beta)}{\beta} \times 100\% \quad \text{Relative bias}$$

Source:

Meeker, W. and Escobar, L, Statistical Methods for Reliability Data, John Wiley & Sons, Inc, New York, 1998.

Nelson, W., *Applied Life Data Analysis*, John Wiley & Sons, Inc., New York, 1982

Proposed Methodology (1/3)



Proposed methodology (2/3)

EXPERT ELICITATION FORM

Name	XX
Position	Engineer
Years at current position	3

Selection of experts based on position and duration at current position. These factors will be included in the calculation of weightage

Subsystem
PUMP

1 List the top 5 failure modes of pumps P201A/B/C

1 Mechanical Seal leak	30 % of total failure
2 High vibration	25 % of total failure
3 Bearing high temperature	20 % of total failure
4 Governor problem	15 % of total failure
5 Lube oil problem	10 % of total failure

Aggregated contribution from each failure modes to the system lifetime

Sum 100 (sum should be 100%)

2 The chance of the following failure modes to occur after running X number of days non-stop operation

	Highly likely (9 out of 10)	50/50 chance	Very unlikely (1 out of 10)	
1 Mechanical Seal leak	180	100	30	Days
2 High vibration	200	100	30	Days
3 Bearing high temperature	200	90	60	Days
4 Governor problem	300	150	100	Days
5 Lube oil problem	300	200	90	Days

Time of occurrence of each failure mode based on 90%, 50% and 10% chance

Proposed Methodology (3/3)

- Governing Equations:

- Distributions:

Weibull

$$f(t) = \frac{\beta}{\eta} \left(\frac{t-\gamma}{\eta} \right)^{\beta-1} e^{-\left(\frac{t-\gamma}{\eta} \right)^\beta}$$

Normal

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2}$$

Lognormal

$$f(t) = \frac{1}{t\sigma'\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t'-\mu'}{\sigma'}\right)^2}$$

$$f(t) \geq 0, t > 0, \sigma' > 0$$

$$t' = \ln(t)$$

- Combination of distributions based on weightage

$$f(x) = w_1 f_1(x) + \dots + w_n f_n(x) \quad w_i > 0$$

$$\sum_{i=1}^n w_i = 1$$

where w_i is the mixing parameter which represents the proportion of mixing of the component distributions. The function $f_i(x)$ is the probability density function of the component distribution i .

- Parameter Estimation using MLE

$$LL = \sum_{i=1}^n \ln(f(x_i))$$

The parameters are solved by equating the derivative of the log likelihood function with respect to each parameters to zero.

Case Study

- The study was to conduct elicitation from experts to predict the life distribution of AGRU pumps.
- 5 experts from various functions were selected:
 - 3 Mechanical Executives
 - 1 RIM rotating
 - 1 Production Supervisor
- The year of service ranging from 0.2 years to 5 years

Case Study: Determining Expert Weightage

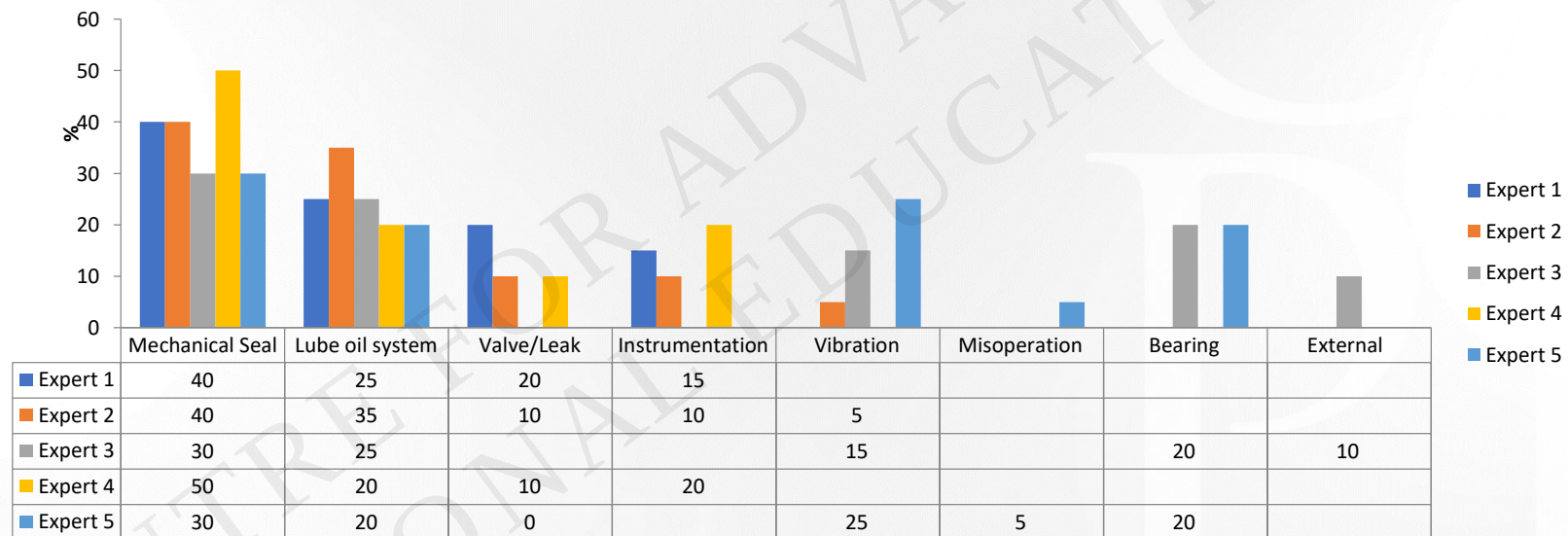
Using Analytic Hierarchy Process (AHP), the weightage for each expert were evaluated.

Experts	Job Function	Years at current function (years)	Weightage
Expert 1	Mechanical Exec	1.5	0.13
Expert 2	Mechanical Exec	3	0.26
Expert 3	RIM Rotating	0.2	0.02
Expert 4	Shift supervisor	4	0.17
Expert 5	Mechanical Exec	5	0.43

Case Study: Failure Modes

Failure mode aggregation determines how each failure mode would contribute to system life distribution

Failure Mode Aggregation



Case Study: Results

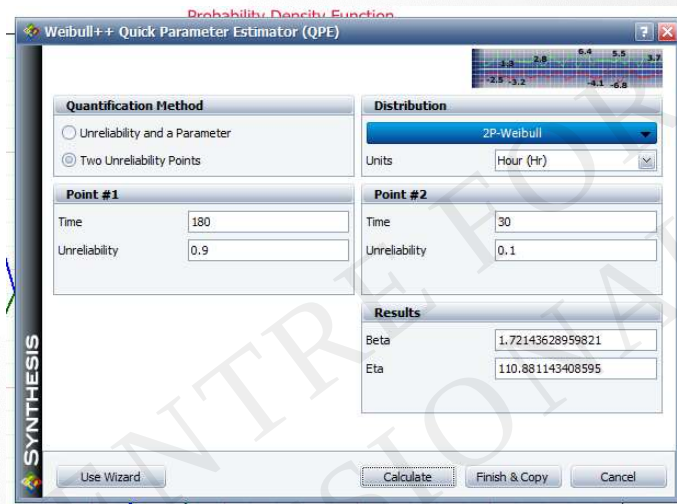
Elicitation results (Samples):

The chance of the following failure modes to occur after running X number of days non-stop operation				
Expert 1				
	Highly likely (9 out of 10)	50/50 chance	Very unlikely (1 out of 10)	
1 SEVERE MECH SEAL LEAK	120	60	30	Days
2 INSTRUMENTATION	200	180	30	Days
3 LUBE OIL PROBLEM	250	180	60	Days
4 MINIMUM FLOW VALVE FAILURE	240	120	60	Days
5				Days

The chance of the following failure modes to occur after running X number of days non-stop operation				
Expert 5				
	Highly likely (9 out of 10)	50/50 chance	Very unlikely (1 out of 10)	
1 Mechanical Seal leak	180	100	30	Days
2 High vibration and noisy	200	100	30	Days
3 Bearing high temperature	200	90	60	Days
4 Lube oil problem	300	200	90	Days
5 misoperation	200	70	70	Days

Case Study: Results

- Estimate distribution for each failure modes:
 - Can be done using Excel built in Solver or Weibull ++ parameter estimation



The image shows the 'Weibull++ Quick Parameter Estimator (QPE)' window. It has a 'Quantification Method' section with two radio buttons: 'Unreliability and a Parameter' (unselected) and 'Two Unreliability Points' (selected). Below this are two sections, 'Point #1' and 'Point #2', each with 'Time' and 'Unreliability' input fields. 'Point #1' has Time: 180 and Unreliability: 0.9. 'Point #2' has Time: 30 and Unreliability: 0.1. A 'Distribution' dropdown is set to '2P-Weibull' with 'Units' set to 'Hour (Hr)'. A 'Results' section shows 'Beta' as 1.72143628959821 and 'Eta' as 110.881143408595. At the bottom are buttons for 'Use Wizard', 'Calculate', 'Finish & Copy', and 'Cancel'.

Weibull ++

Mechanical Seals					
	180	100	30	Weibull Parameters	
Actual	0.884957	0.515859	0.07450376	Beta	1.858332
	0.9	0.5	0.1	Eta	118.8591
Squared Error	0.000226	0.000252	0.000650058		
SSE	0.001128				

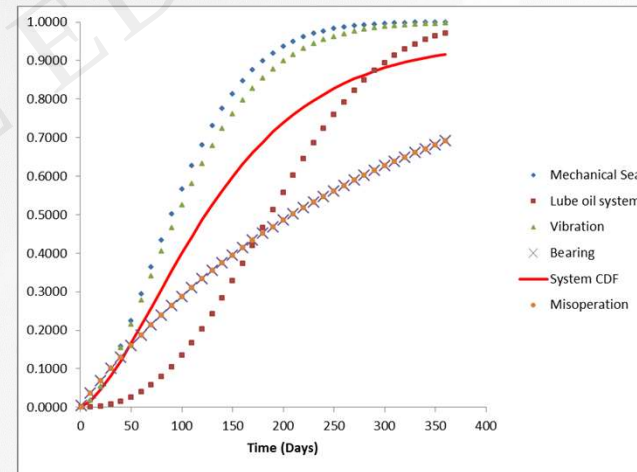
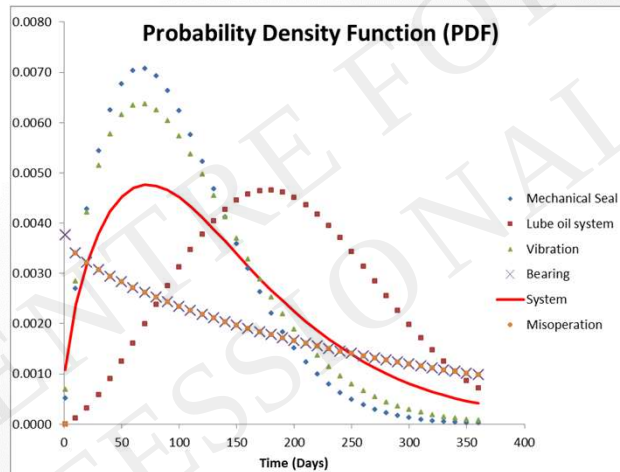
Objective function is to minimized the SSE

Excel's Solver

Case Study: Results

Expert 5

Time (Day)	Mechanical Seal		Lube oil system		Vibration		Bearing		Misoperation		System PDF	System CDF
	PDF1	CDF1	PDF2	CDF2	PDF3	CDF3	PDF4	CDF4	PDF5	CDF5		
1	0.0005	0.0003	0.0000	0.0000	0.0007	0.0004	0.0038	0.0039	0.0000	0.0000	0.0011	0.0010
10	0.0027	0.0158	0.0001	0.0005	0.0028	0.0177	0.0034	0.0357	0.0034	0.0357	0.0024	0.0182
20	0.0043	0.0511	0.0003	0.0026	0.0042	0.0534	0.0032	0.0687	0.0032	0.0687	0.0032	0.0464
30	0.0054	0.1000	0.0006	0.0071	0.0051	0.1004	0.0031	0.1001	0.0031	0.1001	0.0038	0.0815
40	0.0063	0.1587	0.0009	0.0145	0.0058	0.1552	0.0029	0.1301	0.0029	0.1301	0.0042	0.1218
50	0.0068	0.2241	0.0012	0.0252	0.0062	0.2150	0.0028	0.1589	0.0028	0.1589	0.0045	0.1657
60	0.0070	0.2933	0.0016	0.0394	0.0063	0.2777	0.0027	0.1866	0.0027	0.1866	0.0047	0.2119
70	0.0071	0.3640	0.0020	0.0574	0.0064	0.3414	0.0026	0.2133	0.0026	0.2133	0.0048	0.2593
80	0.0069	0.4341	0.0024	0.0792	0.0063	0.4046	0.0025	0.2389	0.0025	0.2389	0.0047	0.3069
90	0.0066	0.5020	0.0028	0.1049	0.0060	0.4661	0.0024	0.2637	0.0024	0.2637	0.0047	0.3540
100	0.0062	0.5664	0.0031	0.1343	0.0057	0.5249	0.0023	0.2875	0.0023	0.2875	0.0045	0.3999

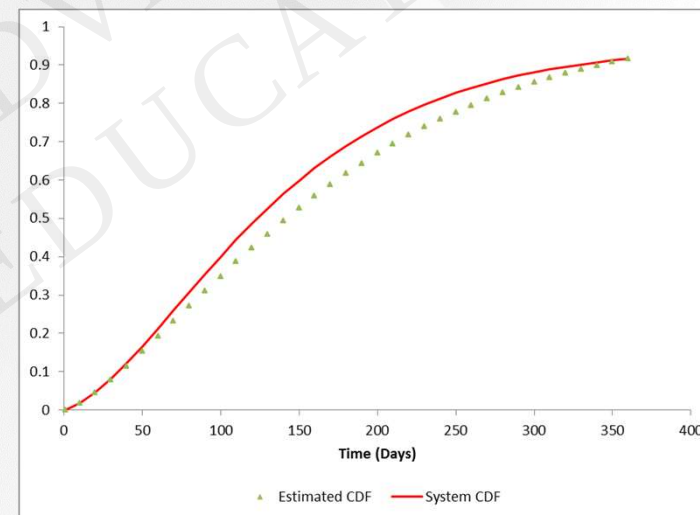
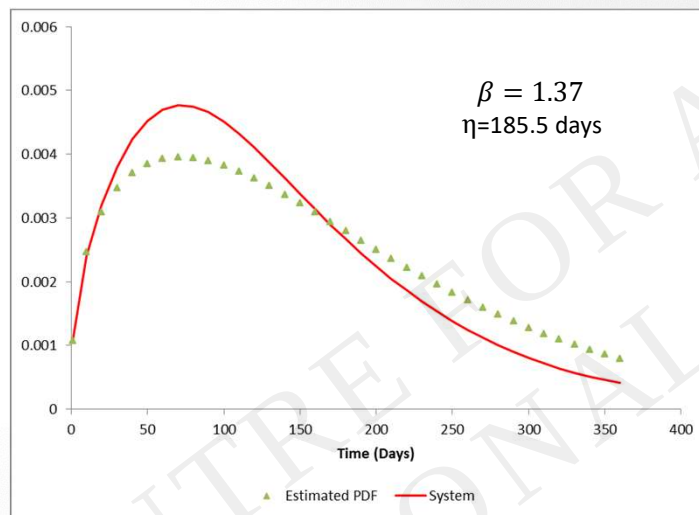


Estimated based on aggregated failures

Case Study: System life distribution

- System life distribution was assumed to follow Weibull distribution
- Parameters of the distribution can be estimated using Excel Solver or Weibull ++ Parameter estimator.

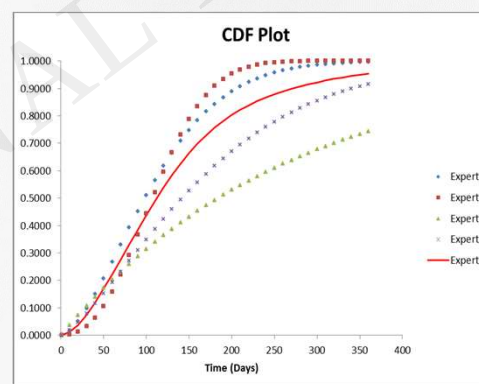
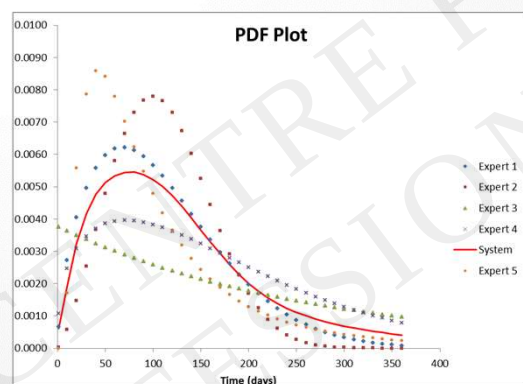
Expert 5



Case Study: Combination of experts

- Similar process as with different failure mode weightage

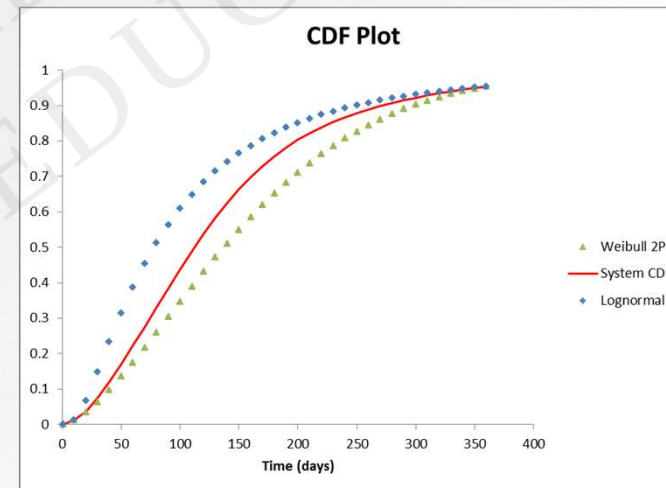
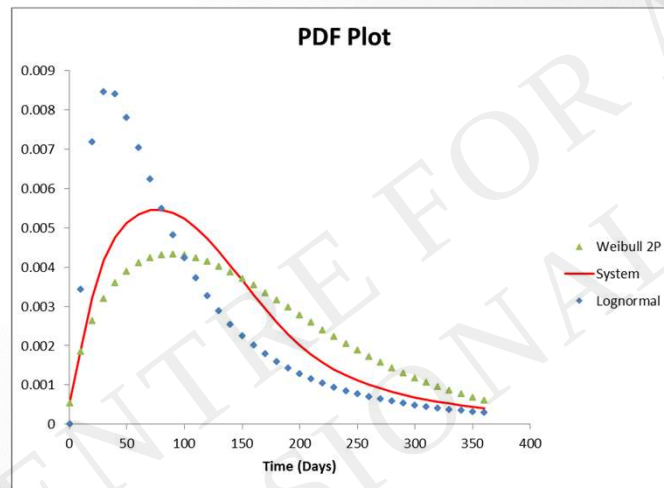
	Expert 1		Expert 2		Expert 3		Expert 4		Expert 5		System PDF	System CDF	
	PDF1	CDF1	PDF2	CDF2	PDF3	CDF3	PDF4	CDF4	PDF5	CDF5			
1		0.0007	0.0004	0.0000	0.0000	0.0038	0.0038	0.0011	0.0008	0.0000	0.0000	0.0006	0.0005
10		0.0027	0.0170	0.0006	0.0024	0.0036	0.0370	0.0025	0.0181	0.0017	0.0047	0.0019	0.0120
20		0.0041	0.0514	0.0015	0.0124	0.0035	0.0727	0.0031	0.0462	0.0056	0.0418	0.0032	0.0379
30		0.0050	0.0967	0.0025	0.0325	0.0034	0.1070	0.0035	0.0791	0.0079	0.1106	0.0042	0.0753
40		0.0056	0.1496	0.0037	0.0636	0.0032	0.1401	0.0037	0.1151	0.0086	0.1938	0.0048	0.1202
50		0.0060	0.2076	0.0048	0.1059	0.0031	0.1719	0.0039	0.1529	0.0084	0.2793	0.0051	0.1698
60		0.0062	0.2684	0.0058	0.1589	0.0030	0.2026	0.0039	0.1919	0.0078	0.3605	0.0053	0.2222
70		0.0062	0.3305	0.0067	0.2213	0.0029	0.2321	0.0040	0.2314	0.0070	0.4347	0.0054	0.2763
80		0.0061	0.3923	0.0073	0.2912	0.0028	0.2606	0.0039	0.2709	0.0062	0.5010	0.0055	0.3308
90		0.0059	0.4528	0.0077	0.3662	0.0027	0.2880	0.0039	0.3101	0.0055	0.5595	0.0054	0.3851
100		0.0057	0.5108	0.0078	0.4438	0.0026	0.3143	0.0038	0.3488	0.0048	0.6108	0.0052	0.4381



Estimated
based on
expert
weightage

System Life Distribution

- Based on the MLE, the system lifetime can be best described by Weibull distribution with parameters:
 - $B = 1.54$
 - $\eta = 173$ days



Result Validation

- Contour plot comparison-To compare life time distribution
- Result show that the system distribution based on elicitation is statistically similar at 90% confidence level

